

Philanthropic characteristics of the Black Hills Area and comparison to selected benchmark regions

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Executive summary

(Need to write it at the end based on conclusions)

1. Introduction

Study background and objectives

The study objective is to describe, through publicly available quantitative data, the philanthropic landscape of the Black Hills region. The Black Hills region is compared to a set of benchmark regions selected to have similar demographic and qualitative characteristics, with the intention of identifying similarities and differences that may inform the strategy of Black Hills Area non-profit institutions.

The Black Hills area of South Dakota is made up of 7 counties in Western South Dakota (Butte, Meade, Lawrence, Pennington, Custer, Oglala Lakota, and Fall River Counties) and a small portion of Eastern Wyoming. This is a largely rural region with a strong tourism industry as well as agriculture, ranching, mining and is the home of Ellsworth Airforce Base. These characteristics were considered when selecting the benchmark regions for comparison, with a goal of identifying benchmark areas with broadly similar demographic, economic, historical, and cultural characteristics.

The philanthropic landscape of the Black Hills region is believed to be driven by its rural character and its tourism-driven economy. This study summarizes demographic characteristics, donor characteristics, and non-profit organization statistics in comparison with benchmark regions, with the intention of identifying similarities, differences, and correlations with philanthropic behavior. The analysis covers the period after the COVID-19 epidemic of 2020- 2021, which represented a significant disturbance in economic activity.

Limitations

General limitations of the study are described here, while limitations specific to individual data sources are described in later sections. This study was restricted to analysis of publicly available data from open-access internet resources rather than being collected specifically to support the desired analyses. Most of the data are self-reported in the context of tax filings and no audits or other accuracy checking

were performed. The desire to focus on the post-covid period meant that some datasets had as few as one year's worth of available data. Multiple years were analyzed where available, as indicated.

Several categories of philanthropic activity are known to be absent from these data due to falling outside the requirements for IRS reporting. For example, crowd-sourced donations such as GoFundMe, and informal donations to family or community members, do not appear in the data sources employed in the report. Donations of any type that are paid by tax filers who don't itemize deductions are not present in the analysis of donor characteristics. Non-itemizing filers constitute a large majority of taxpayers in the Black Hills and benchmark regions, so this is an especially noteworthy limitation. Similarly, non-profit organizations who are exempt from IRS reporting due to their size or function (ie, certain religious organizations) are not present in the organization-centered datasets and may make up a large proportion of non-profit organizations.

Correlations between demographic characteristics, donor behavior, and other variables may be identified, but causal relationships cannot be inferred.

2. Data sources

What data sources were used and why?

What are their limitations?

Note appendices for details.

GivingTuesday 990 DataMart

The GivingTuesday 990 DataMart is the main source used for non-profit financial information in this report. It is based on public IRS filings submitted by tax-exempt organizations, including Form 990, Form 990-EZ, and Form 990-PF. These forms report information such as total revenue, contributions, program service revenue, expenses, assets, filing year, form type, and organization identifiers.

The IRS form type matters because different organizations file different versions of the Form 990 family. Form 990 is the full annual return for larger non-profit organizations and provides the most detail. Form 990-EZ is a shorter return used by some smaller organizations and includes less detail. Form 990-PF is used by private foundations. Form 990-N, also called the e-Postcard, is used by very small organizations and contains only limited information, so those organizations are not included in the GivingTuesday financial analysis used here.

The project uses the GivingTuesday Basic Fields files for Form 990, Form 990-EZ, and Form 990-PF, limited to organizations located in the Black Hills and benchmark-region counties. When GivingTuesday did not include all classification details needed to group organizations consistently, the project supplemented those details with public registry sources from NCCS and the IRS.

The main strength of the GivingTuesday source is that it provides more financial detail than registry-only sources. This makes it useful for describing organization size, revenue sources, and overall financial composition. However, it should not be read as a complete count of every non-profit or every charitable activity in a region. Very small organizations that file only Form 990-N, organizations not required to file public Form 990-family returns, houses of worship and some related organizations, and informal giving are not fully represented.

The source also has limits because the IRS forms do not all ask for the same level of detail. Organizations filing the full Form 990 report more detail about where contributions come from. Organizations filing Form 990-EZ or Form 990-PF report less detail in some areas. In addition, one Form 990 contribution line combines several donor types, including individual gifts, foundation grants, donor-advised fund distributions, corporate gifts, bequests, and other contributions. Because of this, the data cannot cleanly split every contribution dollar into individual giving versus institutional giving.

For this reason, the report uses GivingTuesday as a practical view of reported non-profit finances on IRS filings. It does not capture all generosity in the region. The results are strongest when they describe broad regional patterns rather than when they try to assign every dollar to a specific donor type.

The appendices provide additional detail on the source files and fields used.

American Community Survey (ACS)

The American Community Survey (ACS) is an ongoing U.S. Census Bureau program that surveys approximately 3.5 million households annually to provide up-to-date information on social, economic, housing, and demographic characteristics. For this report, ACS data provide county-level information on household income, employment, and civilian and armed forces composition from 2022 to 2024.

The use of ACS data in this report is twofold. First, ACS data provide the economic and demographic information needed to assess whether, and to what extent, the Black Hills region is comparable to other regions, thereby supporting data-driven benchmark region

selection. Second, ACS data contribute to the analysis of factors that may explain differences in the nonprofit organizational landscape between the Black Hills region and the benchmark regions.

One limitation of ACS data is that only housing units with long-term residents are eligible for inclusion. The ACS uses the concept of “current residence” to determine who should be counted within sampled housing units. Specifically, individuals who live at an address for less than two months in a year are not included in the survey. As a result, in regions with a high proportion of vacation homes, seasonal residents in those homes are not captured in ACS estimates.

The ACS appendix provides additional details on data provenance, extraction, and processing, as well as descriptions of the specific data elements extracted.

Bureau of Labor Statistics Non-profit Research Dataset

The Bureau of Labor Statistics (BLS) provides specialized research data on U.S. non-profit employment, wages, and establishments, with a primary focus on 501(c)(3) organizations. The dataset used in this report contains 2022 data. BLS data provide an overview of how employment and wages in the nonprofit sector compare with the broader private sector across multiple geographic levels, including counties, metropolitan statistical areas, and states.

There are several limitations to the BLS data. First, only 2022 data are available, which prevents examination of changes in the nonprofit sector over time. Second, for privacy reasons, BLS suppresses county-level estimates in areas with a small number of private establishments. As a result, county-level data are unavailable for five counties across the Black Hills region and the benchmark regions. Third, only organizations classified as 501(c)(3) are included in the dataset, which limits the extent to which findings can be generalized to nonprofit organizations with other tax-exempt classifications.

The BLS appendix provides additional details on data provenance, extraction, and processing, including the specific counties excluded from the analysis and definitions of the variables used.

Geographic Area

Land area measures from the U.S. Census Bureau provide context for comparing the size of the Black Hills region with selected benchmark regions and for estimating population density. The land area measure excludes water bodies within census boundaries.

The geographic area appendix provides additional details on data extraction, aggregation, and transformation.

3. Selection and demographic analysis of benchmark regions

Benchmark region selection

The primary objective of the benchmark region selection process was to assess the set of regions that had been pre-identified by the client, for reasonable comparability, and to verify that they closely resemble the Black Hills in key structural characteristics. These characteristics include population size, economic composition, rural–urban balance, and income levels. More specifically, the selection aimed to:

- ensure comparability in population scale, targeting regions with populations broadly within the range of approximately 75,000 to 155,000 at the county level.
- match rural and urban composition, using census definitions to evaluate the proportion of urban versus rural areas.
- capture similar economic structures, and align demographic and socioeconomic indicators, such as age distribution, employment patterns, and income levels.

Overall, the goal was to ensure that the set of benchmark regions suggested by the client provide meaningful context for comparison by reflecting similar underlying economic and demographic conditions.

The eventually selected benchmark regions, **Sioux Falls**, which is made up of 5 counties (Minnehaha, Lincoln, McCook, Turner, and Rock), **Billings**, which is made up of 3 counties (Carbon, Stillwater, Yellowstone), **Flagstaff**, which has 1 county (Coconino), and **Missoula**, which is made up of 2 counties (Mineral, Missoula), offer a balanced comparison set. Together, these regions establish a robust foundation for analyzing how the Black Hills compares to peer regions.

Relevant details about data sources

The demographic analysis of the Black Hills and benchmark regions relied on data from the American Community Survey (ACS), which provides annually updated estimates based on survey samples rather than complete population counts. For this study, the latest ACS 5-year estimates were used due to their full geographic coverage and higher reliability for smaller and rural regions. These estimates aggregate data over a five-year period, producing a single, stable value for each variable.

While this improves statistical reliability, it also means that the data represent average conditions over time rather than for specific years, limiting the ability to capture short-term changes or trends.

Additionally, because ACS outputs are pre-aggregated estimates, each region is represented by a single value per variable, which restricts the use of inferential statistical testing. As a result, the demographic comparison across benchmark regions is interpreted using descriptive analysis and visual comparison, focusing on relative magnitudes and patterns rather than statistically significant differences. This approach ensures that comparisons remain meaningful while acknowledging the limitations inherent in the data structure.

Results of benchmark regions analyses

This section presents a comparative analysis of the demographic characteristics of the Black Hills region in relation to the selected benchmark regions. Analyses were carried out across key structural characteristics, including population size, labor market dynamics, occupational composition, and household income.

Population Size and Structure

The Black Hills region has a smaller total population compared to the Sioux Falls region but has a higher total population than the other three benchmark regions (Figure 1a).

However, the population aged 16 and above in the Black Hills region shows a substantial proportion of the total population falling within the working age category, which is consistent with the patterns observed in the benchmark regions (Figure 1b).

Figure 1a: Total Population of Black Hills with Bench Mark Regions
(ACS 5-Year Estimate Data, 2024)

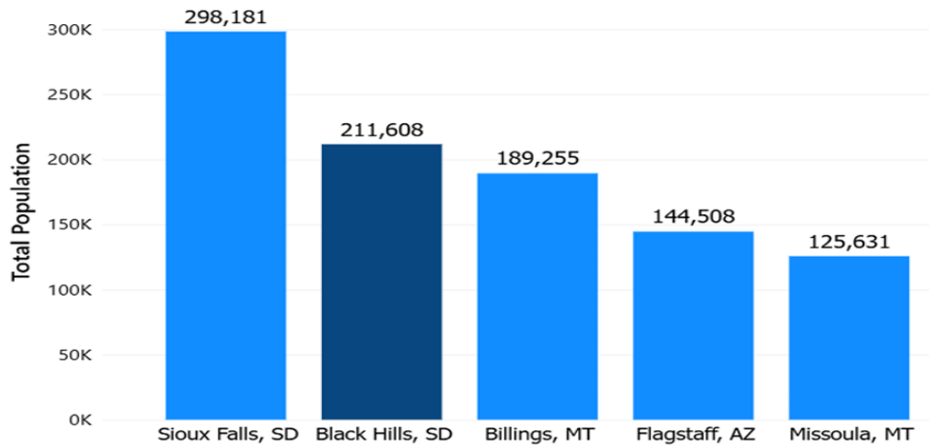
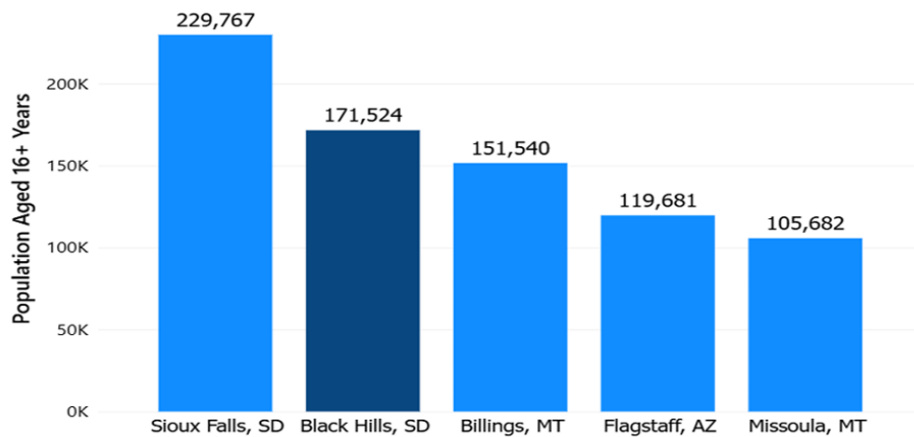


Figure 1b: Population (Aged 16+Years) of Black Hills with Bench Mark Regions
(ACS 5-Year Estimate Data, 2024)



Labor Market Characteristics

Considering the population aged 16 and above, a larger fraction of them is in the labor force, across all the regions, while the smaller fraction is categorized as not in labor force (Figure 2a). The labor force category is also classified into Civilian labor force, which takes the majority, and the Military labor force which takes only a very small, negligible share. The labor force signifies individuals aged 16 and above that are either employed or actively seeking employment.

Within the civilian labor force, employment levels are consistently high across both the Black Hills and benchmark regions, with unemployment levels representing a relatively small proportion (Figure 2b).

Overall, the Black Hills region demonstrates labor market participation patterns that are broadly comparable to the benchmark regions, indicating similar levels of workforce engagement, despite differences in population size.

Figure 2a: Population (Aged 16+Years) that are In labor force and those not in labor force, in Black Hills and Bench Mark Regions

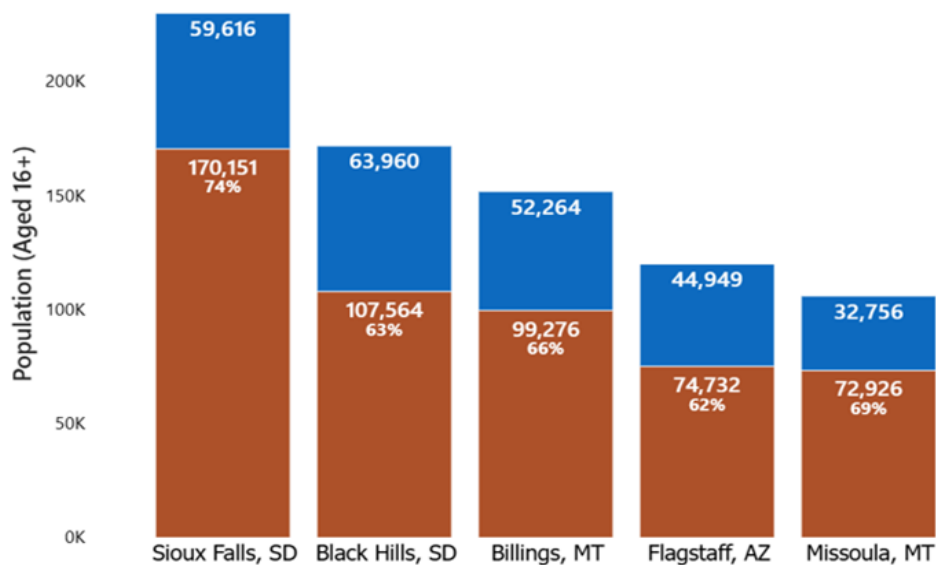
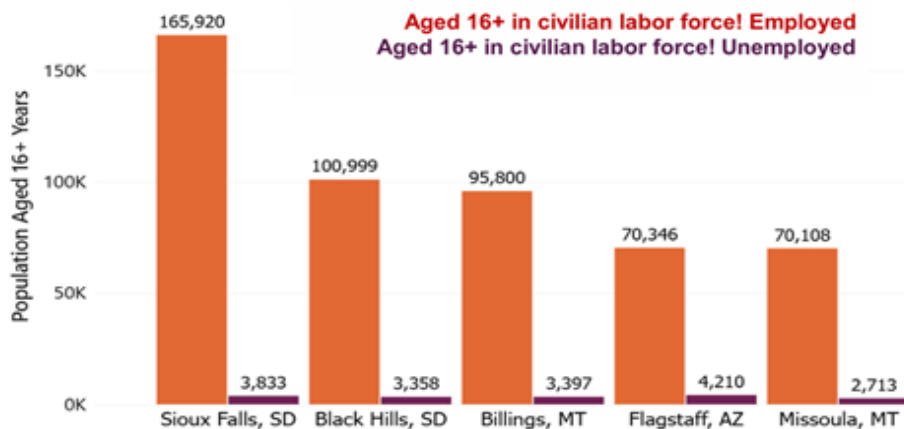


Figure 2b: Population (Aged 16+Years in civilian labor force) that are employed and those not employed, in Black Hills and Bench Mark Regions



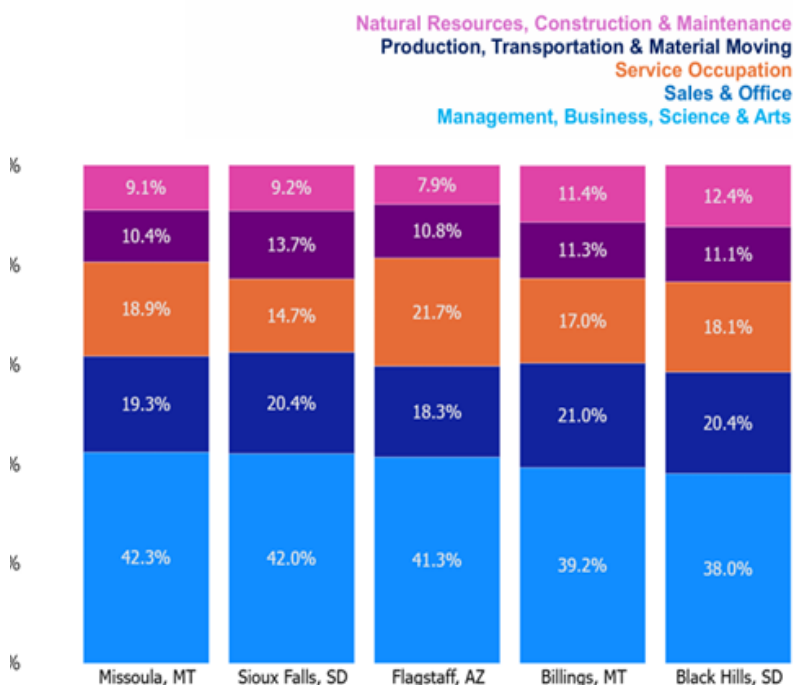
Occupational Structure

The occupational distribution highlights how employment is spread across major sectors. In the Black Hills and benchmark regions, employment is distributed across five major sectors which are: Natural resources, construction, and maintenance; Production, transportation, and material moving; Service occupations; Sales and office occupations; and Management, business, science, and arts occupations.

The occupational distribution in the Black Hills and benchmark regions reflect notable representation in all five sectors. Although, compared to benchmark regions (Figure 3), the Black Hills shows a slightly higher concentration in the natural resources, construction and maintenance occupations, with a slightly lower proportion of employment in the management, business, science, and arts occupations, while the benchmark regions tend to have a slightly larger proportion of employment in management, business, science, and arts occupations.

However, the general overview demonstrates comparable occupational distribution patterns across all regions.

Figure 3: Occupational Distribution of Civilian Employed Aged 16+ years in Black Hills Compared with Benchmark Regions



Household Income Characteristics

The visualization of total households (Figure 4a) indicates that the Black Hills has fewer households than Sioux Falls but higher than the other benchmark regions, which is consistent with the population size visualization (Figure 1a).

A notable feature is the presence of households with retirement income, which constitutes an important segment of the population (Figure 4b). The Black Hills shows a measurable share of households receiving retirement income, suggesting the presence of an older or retired population segment (Figure 4c).

Figure 4a: Total Number of Households in Black Hills with Bench Mark Regions

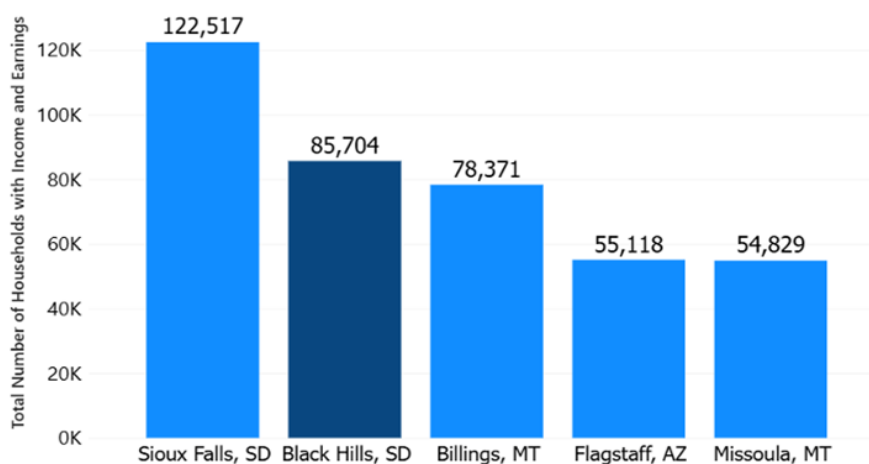


Figure 4b: Total Number of Households with Retirement Income in Black Hills and Bench Mark Regions

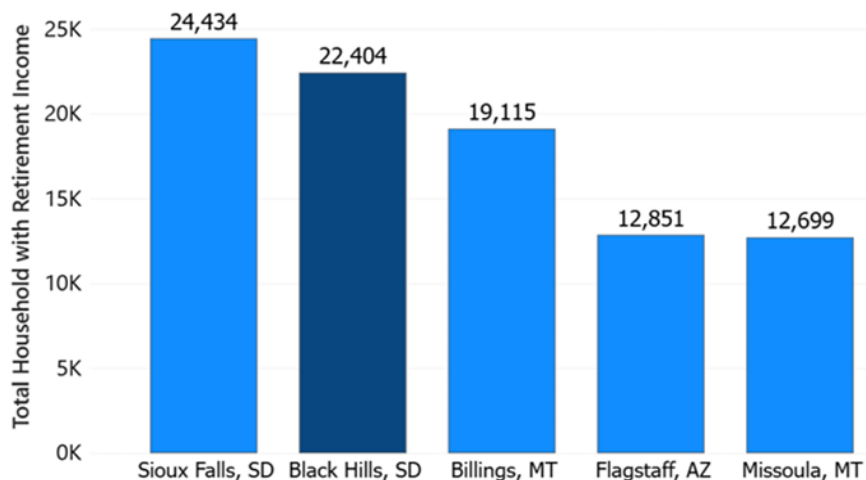
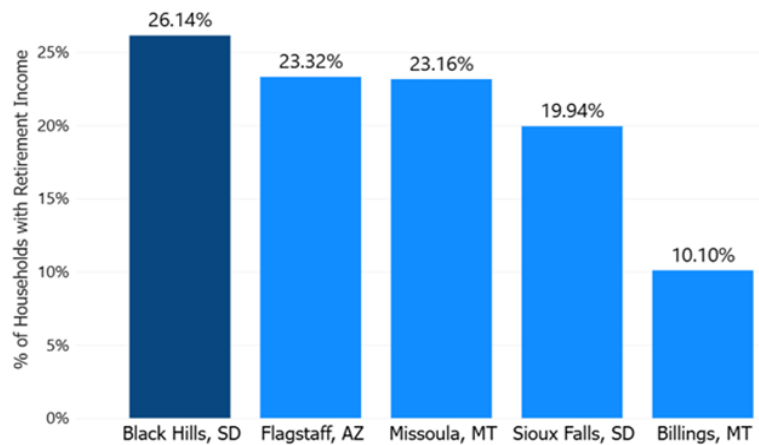


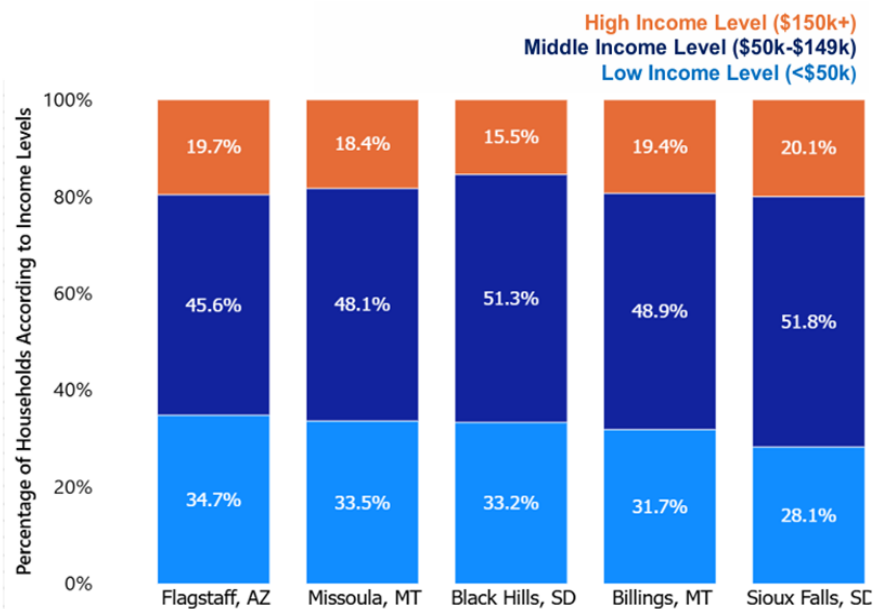
Figure 4c: Percentage of Households with Retirement Income in Black Hills and Bench Mark Regions



Income Distribution

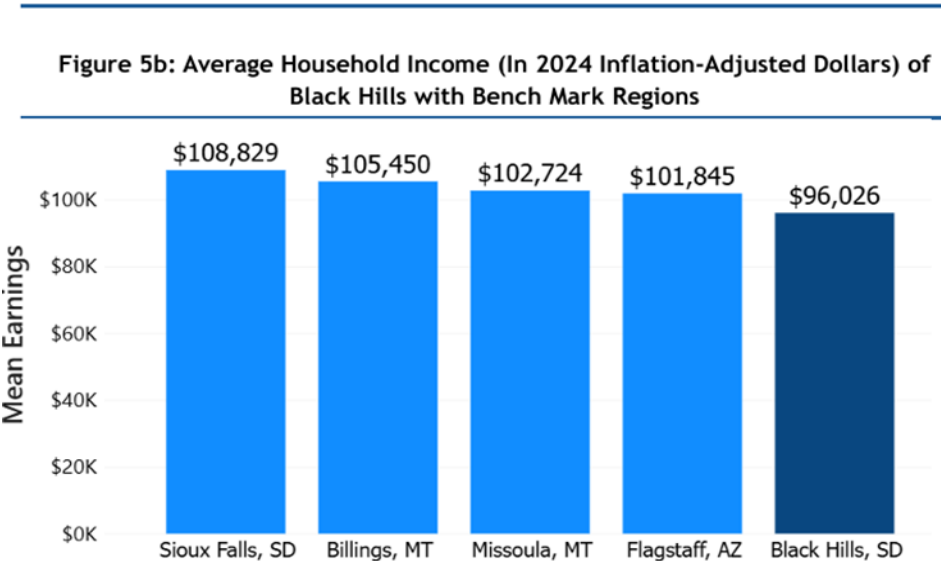
Household income distribution is divided into three categories: low (<\$50,000), middle (\$50,000–\$149,000), and high (\$150,000+) income. Although the Black Hills region tends to reflect a slightly lower share in the high-income category, the general household income distribution overview demonstrates comparable income distribution patterns across all regions (Figure 5a).

Figure 5a: Household Income Distribution (%) in Black Hills Compared with Bench Mark Regions



Average Household Income

Average household income in the Black Hills, adjusted for inflation, provides a key indicator of overall economic well-being. While the Black Hills region demonstrates stable income levels, benchmark regions report slightly higher average household incomes (Figure 5b). This disparity may suggest a comparatively smaller pool of high-capacity donors in the Black Hills region.



4. Characterization of donation activity

This section describes tax-reported donor behavior in the Black Hills region relative to the four benchmark regions, with attention to how charitable giving varies across regions once differences in income are taken into account. The analysis draws on county-level tax data and characterizes reported giving patterns.

Relevant details about data sources

The analysis in this section uses the Internal Revenue Service Statistics of Income (SOI) 2022 County Data. Regions are defined by aggregating the counties that make up each area. All dollar values in the source file are reported in thousands, and the figures shown here have been converted to whole dollars.

Returns are grouped into three income bands based on adjusted gross income: low (under \$50,000), middle (\$50,000 to \$100,000), and high (\$100,000 or more). Two features of the data shape how the results should be read. First, charitable contributions appear in tax data only for filers who itemize their deductions, so giving by non-itemizers is not observable. Because non-itemizers are a large majority of filers in these regions, observed giving will understate total giving. Second, the data are totals at the county-level and do not identify individual donors, so they cannot distinguish one-time from recurring gifts, account for household size, or describe the relationship between donors and the organizations they support.

Results of donation activity analysis

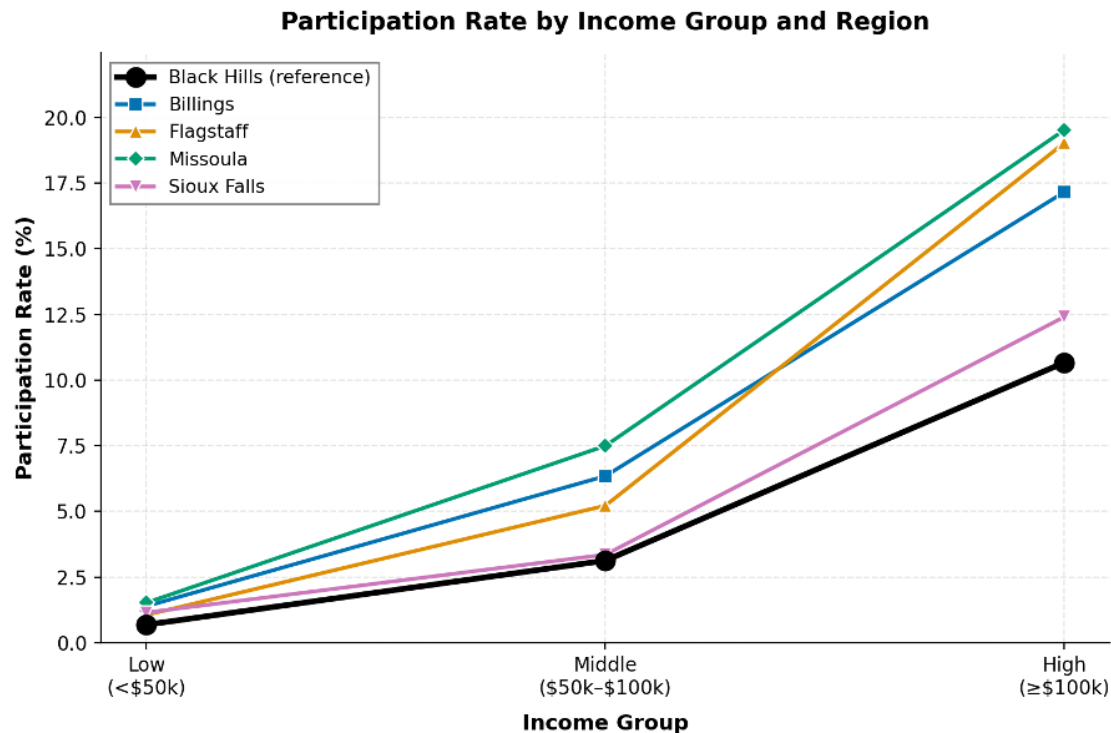
Donor participation

Donor participation is measured as the share of tax returns that report charitable contributions out of all returns filed. Participation rises with income in every region. Across all income groups, the Black Hills region has a lower participation rate than each benchmark region, both overall (3.5 percent) and within each income band.

Table 4-1. Donor Participation Rate (%) by Income Group and Region.

Income group	Black Hills	Billings	Flagstaff	Missoula	Sioux Falls
Low (<\$50k)	0.7	1.4	1.1	1.5	1.2
Middle (\$50k–\$100k)	3.1	6.4	5.2	7.5	3.4
High (≥\$100k)	10.7	17.2	19.0	19.5	12.4
Region overall	3.5	6.4	5.7	6.9	4.7

Figure 4-1. Donor Participation Rate by Income Group and Region



Itemization and Observed Participation

Donor participation also relates to itemization, since only filers who itemize can report charitable deductions, and the itemization rate therefore sets the range within which giving can be observed. The Black Hills region has the lowest itemization rate of the five regions, at 4.6 percent. Among filers who do itemize, the share who report charitable gifts is similar across regions, between about 76 and 84 percent. This points to the region's lower observed participation being tied more to how few filers itemize than to differences in reported giving among those who do. The same pattern holds within each income band. Because state tax rules can affect the incentive to itemize deductions, regional differences in how many filers itemize, and therefore in observed participation, may partly reflect differences in local tax policy. This is a contextual consideration only and was not examined in the present analysis.¹

Table 4-2. Participation, Itemization, and the Participation-to-Itemization Ratio by Region (%)

Measure	Black Hills	Billings	Flagstaff	Missoula	Sioux Falls
Participation	3.5	6.4	5.7	6.9	4.7
Itemization	4.6	8.4	7.1	9.1	5.6

¹State tax provisions reflect rules in effect for the 2022 tax year, the year of the IRS SOI data used in this analysis. South Dakota levies no state individual income tax. Montana allowed charitable contributions as itemized deductions, subject to state-specific exceptions, while Arizona provided nonrefundable individual income tax credits for cash contributions to qualifying charitable organizations and qualifying foster care charitable organizations certified by the Arizona Department of Revenue. Sources: South Dakota Department of Revenue (dor.sd.gov); Montana Department of Revenue, 2022 Form 2 Instructions (Itemized Deductions Schedule); Arizona Department of Revenue, credits for contributions to QCOs/QFCOs (azdor.gov).

Participation / Itemization	76.1	75.7	80.8	76.1	83.8
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Table 4-3. Itemization Rate (%) by Income Group and Region

Income group	Black Hills	Billings	Flagstaff	Missoula	Sioux Falls
Low (<\$50k)	1.4	2.2	1.6	2.5	1.6
Middle (\$50k–\$100k)	4.4	9.5	7.0	10.6	4.6
High (≥\$100k)	12.7	20.8	22.2	23.7	13.8
Region overall	4.6	8.4	7.1	9.1	5.6

Average Contribution per Donor

Average annual contribution per donor is total charitable contributions divided by the number of returns that report them. This reflects how much participating donors give rather than the size of any single gift, since the data report annual totals per return rather than individual donations.

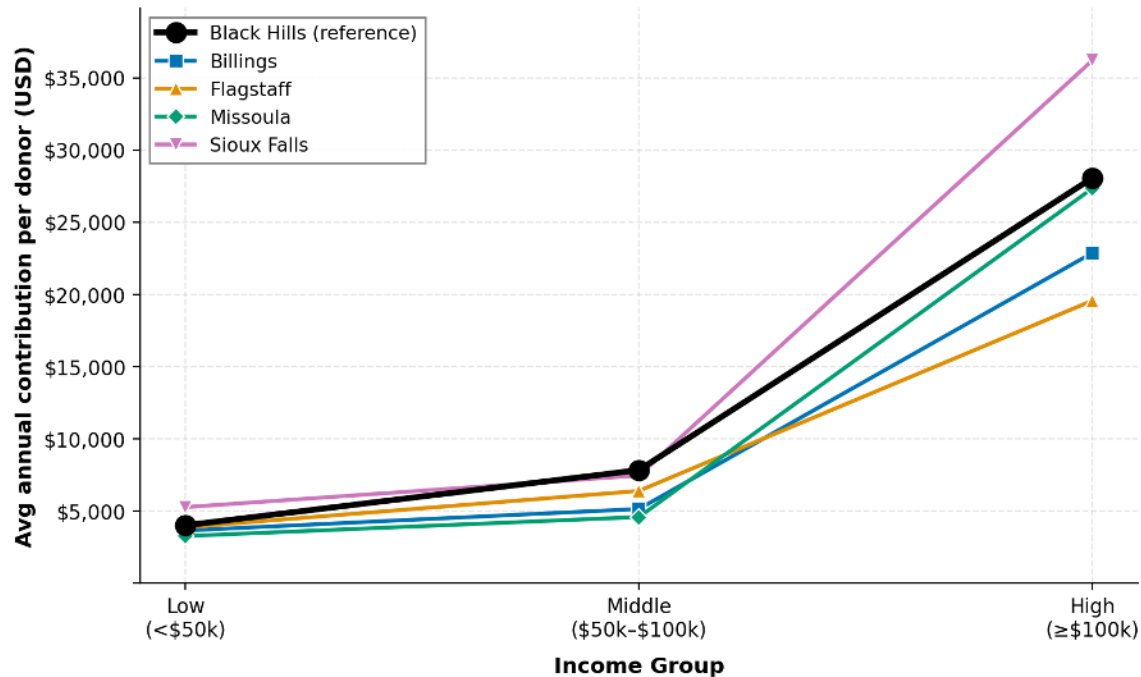
Among participating donors, Black Hills gives competitively across all income groups. Black Hills ranks first in the Middle-income group and second in the Low- and High-income groups, behind only Sioux Falls. Overall, Black Hills ranks second among the five regions, again behind Sioux Falls and well above the other three benchmarks.

Table 4-4. Average Annual Charitable Contributions per Donor by Income Group and Region, among returns reporting contributions

Income group	Black Hills	Billings	Flagstaff	Missoula	Sioux Falls
Low (<\$50k)	\$3,997	\$3,639	\$3,865	\$3,259	\$5,260
Middle (\$50k–\$100k)	\$7,826	\$5,132	\$6,386	\$4,574	\$7,446
High (≥\$100k)	\$28,061	\$22,871	\$19,566	\$27,344	\$36,212
Region Overall	\$21,014	\$16,337	\$15,096	\$18,497	\$27,193

Figure 4-2. Average Annual Charitable Contributions per Donor by Income Group and Region, among returns reporting contributions

Average Annual Charitable Contribution per Donor by Income Group and Region



Giving Relative to Income

Two measures relate giving to income. The region-wide measure divides total contributions by the total adjusted gross income of all taxpayers, which shows the share of a region's income that goes to charity. The itemizer-conditional measure divides total contributions by the estimated adjusted gross income of itemizers only, which shows the share of itemizers' income that is deducted as donations. The two measures place the Black Hills region quite differently. Under the region-wide measure (Panel A) it is the lowest in every income group. Under the itemizer-conditional measure (Panel B) it moves toward the middle or top of the range, ranking first in the middle-income group and second overall. The contrast shows that the low region-wide ratio has more to do with how many filers report giving than with how much itemizers actually give.

Table 4-5. Giving ratio (%) by Income Group and Region

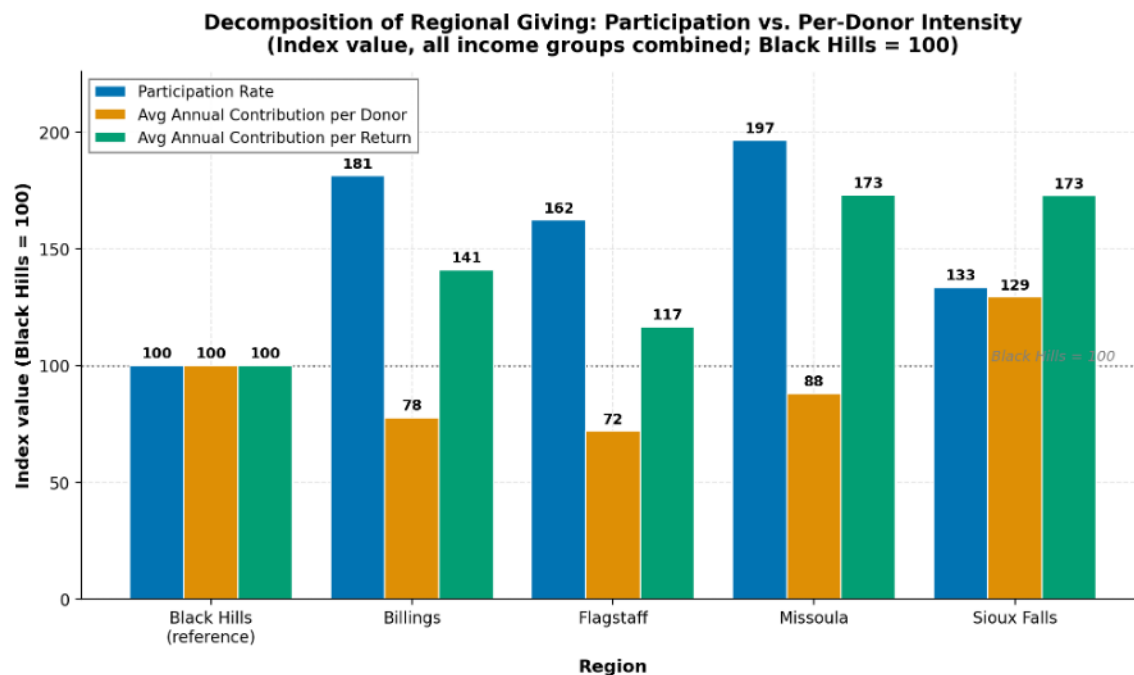
Income group	Black Hills	Billings	Flagstaff	Missoula	Sioux Falls
Panel A. Region-wide (all returns)					
Low (<\$50k)	0.13	0.23	0.18	0.23	0.26
Middle (\$50k–\$100k)	0.34	0.45	0.47	0.48	0.35
High (≥\$100k)	1.32	1.76	1.60	2.09	1.69
Panel B. Itemizer-conditional					
Low (<\$50k)	7.33	8.66	9.38	7.15	12.45

Middle (\$50k–\$100k)	7.67	4.57	6.44	4.39	7.42
High (≥\$100k)	7.09	6.53	5.48	6.75	7.74

Participation versus Contribution Size

Differences in total giving per taxpayer reflect two components: the share of taxpayers who donate (participation) and the amount each donor gives (average contribution per donor). Because giving per return is the product of these two figures, the gap between each benchmark region and the Black Hills region can be attributed to one or the other. Indexed to the Black Hills region at 100, every benchmark region is higher on both participation and giving per return, while the average contribution per donor is comparable or lower in three of the four. The differences in total giving therefore come mainly from participation, not from how much each donor gives. Black Hills donors give as much as or more than donors elsewhere, and the region's lower overall giving traces back to the smaller share of people who give.

Figure 4-3. Decomposition of Regional Giving into Participation and Per-Donor Intensity



Rurality and Donor Participation

Donor participation can also be compared to how rural each county is. Rurality is measured with the USDA Rural-Urban Continuum Code (RUCC), which ranges from 1 (large metropolitan areas of more than one million people) to 9 (small rural counties not adjacent to any metropolitan area).

Across the 18 counties, the rank correlation between rurality and participation is negative but weak ($r_s = -0.28$, $p = 0.26$), consistent with somewhat lower participation in more rural counties. Here, r_s is the Spearman rank-correlation estimator (an estimate of the population parameter ρ); it ranges from -1 to $+1$, with values near zero indicating little monotonic association, and p is the probability of observing a correlation at least this strong if rurality and participation were in fact unrelated. Given the small number of counties, the analysis has limited power to distinguish a

modest association from none, and the magnitude and direction of the coefficient are therefore more instructive than its position relative to a fixed significance threshold.

This association should not be read as causal. Because more rural counties also tend to have lower incomes, and lower-income filers are less likely to itemize and report charitable deductions, the apparent rurality-participation link may instead reflect underlying differences in income rather than rurality itself.

A sensitivity check excluding Oglala Lakota County indicates that the negative association is strongest in the low-income group and persists after the exclusion. As one of several correlations examined, however, this result warrants confirmation with a larger set of counties.

Figure 4-4. Donor Participation Rate against rurality (RUCC) for the 18 study counties

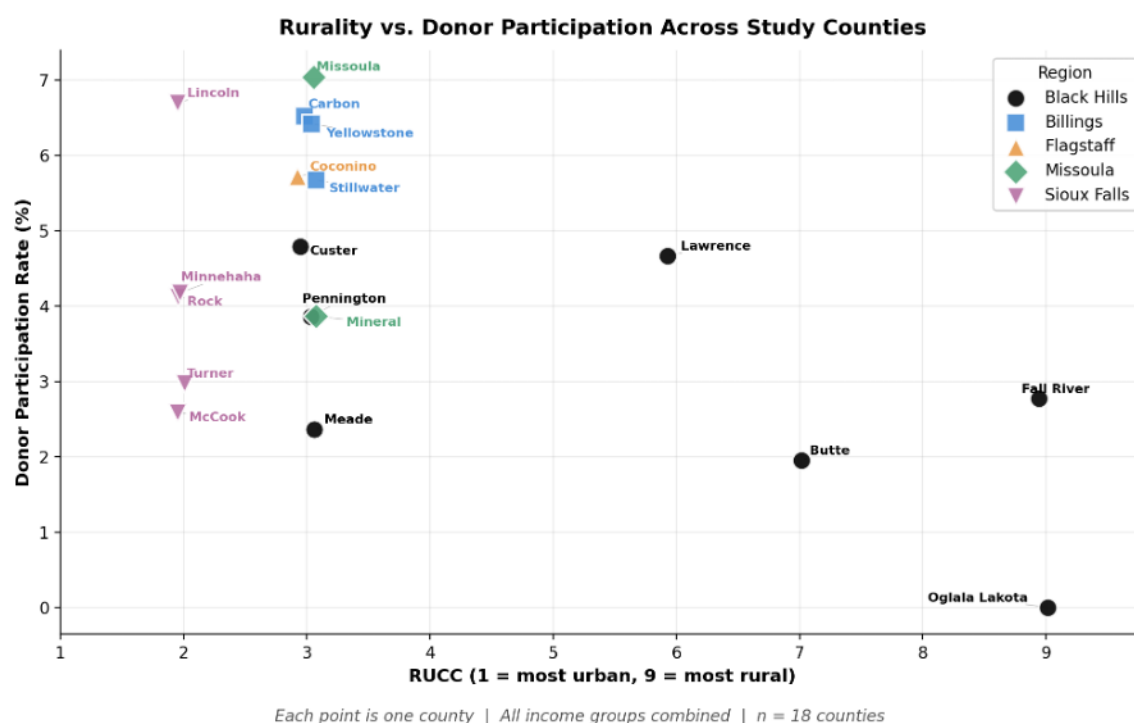


Table 4-6. Spearman rank correlation between rurality (RUCC) and donor participation rate

Income group	Full sample (n = 18)	Excl. Oglala Lakota (n = 17)
All incomes	$r_s = -0.28$ (p = 0.26)	$r_s = -0.14$ (p = 0.61)
Low (<\$50k)	$r_s = -0.67$ (p = 0.003*)	$r_s = -0.61$ (p = 0.009*)
Middle (\$50k–\$100k)	$r_s = +0.17$ (p = 0.51)	$r_s = +0.36$ (p = 0.16)
High (≥\$100k)	$r_s = -0.11$ (p = 0.66)	$r_s = +0.07$ (p = 0.81)

r_s = Spearman rank-correlation coefficient; p = two-sided significance; * denotes $p < 0.05$

Nonprofit Density and Donor Participation

To assess whether the local availability of organizations is related to giving, donor participation is compared with the supply of nonprofits across the same 18 counties. Nonprofit density is the

number of tax-exempt organizations registered with the IRS per 10,000 residents, drawn from the 2022 IRS Exempt Organizations Business Master File and 2022 American Community Survey County population estimates.

Across all 18 counties, nonprofit density shows essentially no relationship with overall donor participation (Pearson $r = 0.07^2$, $p = 0.77$), and the pattern is consistent within income groups (Table 4-6). In this case the coefficient is itself near zero, indicating little apparent association rather than a relationship obscured by the limited sample.

Table 4-7. Pearson Correlation between Nonprofit density and Donor Participation Rate

Income group	Excl. hospitals / univ. / political	All included
All incomes	$r = 0.07$ ($p = 0.77$)	$r = 0.06$ ($p = 0.81$)
Low (<\$50k)	$r = -0.09$ ($p = 0.72$)	$r = -0.11$ ($p = 0.67$)
Middle (\$50k–\$100k)	$r = +0.36$ ($p = 0.14$)	$r = +0.37$ ($p = 0.14$)
High ($\geq$ \$100k)	$r = +0.005$ ($p = 0.99$)	$r = -0.009$ ($p = 0.97$)

Summary

Black Hills' lower overall donation rate appears to be driven primarily by participation, not by lower giving amounts among donors. Once the analysis is limited to itemizers, the participation gap largely narrows, suggesting that itemization behavior is an important part of the observed regional difference.

Across the five regions, the Black Hills region has the lowest donor participation rate (3.5 %) and the lowest itemization rate (4.6 %). Among filers who itemize, the share reporting charitable contributions is comparable to the benchmark regions, ranging from 76% to 84%. Average contributions per donor are also comparable to or higher than the benchmark regions, including the highest value in the middle-income group. The two giving-relative-to-income measures place Black Hills differently: relative to total regional income, it gives the smallest share in every income group but relative to itemizers only, it ranks first in the middle-income group and second overall. Together, these results suggest that the gap in total giving between Black Hills and the benchmark regions reflects differences in participation rather than in the amount each donor gives.

Rurality shows a weak negative association with participation, more pronounced in the low-income group, though the small number of counties limits what can be concluded, and the pattern is not interpreted as causal. Nonprofit density shows little or no association with participation, with correlations close to zero across income groups.

² Both r and r_s are correlation coefficients: r (Pearson) measures linear association between continuous variables, while r_s (Spearman) measures monotonic association based on ranks and is used in the rurality analysis above for ordinal data such as RUCC. The two are labeled Pearson and Spearman throughout.

5. Characterization of non-profit organizations

Relevant details about data sources

Anything the reader needs to know about how the data sources, availability, etc impact the interpretation of this section

Non-profit revenue analysis

This section uses the GivingTuesday Form 990-family data described under Data sources (coverage limits, form types, and what is not in the file).

For tax year 2022, after requiring positive total revenue, the analysis file includes 1,799 filing records for 1,797 organizations: 422 in the Black Hills and 1,377 in benchmark counties. The filing mix is 1,085 Form 990 records, 563 Form 990-EZ records, and 151 Form 990-PF records. Hospitals, universities, and political organizations are excluded for peer comparability (25 records removed from the 2022 universe).

The revenue sources analyzed in this section are defined below.

Revenue source	What it measures	How it is reported on IRS filings
Total revenue	All reported revenue for the organization in tax year 2022.	Form 990, Form 990-EZ, and Form 990-PF totals.
Total contributions	Total reported contributions, gifts, grants, and similar amounts.	Form 990, Form 990-EZ, and Form 990-PF totals.
Program service revenue	Money earned from services, programs, fees, or sales related to the organization's mission.	Form 990 and Form 990-EZ; not comparable on Form 990-PF.
Government grants received	Funds reported from government sources.	Form 990 Part VIII Line 1e; pairwise tests use Form 990 filers only.
Federated campaign contributions	Support reported through federated campaigns (for example, United Way-style allocations).	Form 990 Part VIII Line 1a; pairwise tests use Form 990 filers only.
Related organization contributions	Contributions from related organizations or affiliates.	Form 990 Part VIII Line 1d; pairwise tests use Form 990 filers only.
Membership dues	Membership dues reported on the return.	Form 990 Part VIII Line 1b; pairwise tests use Form 990 filers only.
Fundraising event contributions	Contributions tied to fundraising events.	Form 990 Part VIII Line 1c; pairwise tests use Form 990 filers only.

Mixed / unclassified contributions	Contribution dollars the forms do not break down cleanly by donor type.	Form 990 Line 1f plus 990-EZ/PF contribution totals that cannot be split into Part VIII lines.
Other revenue	Remaining revenue after program service revenue and the contribution components above.	Calculated as total revenue minus those components.

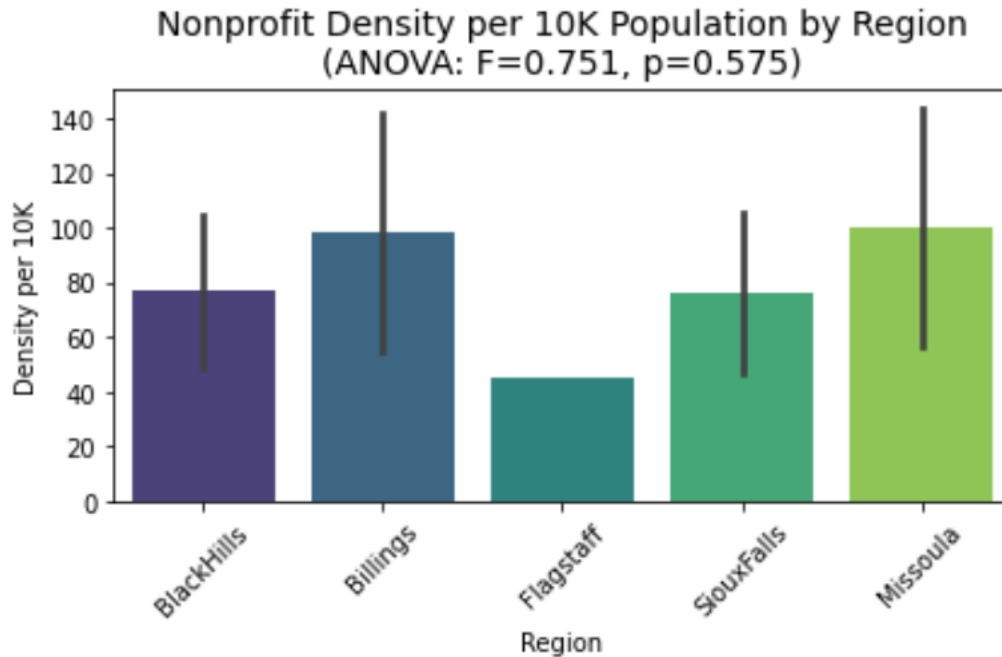
The data cannot cleanly separate individual giving from foundation grants, corporate gifts, bequests, or other institutional support. That limitation applies especially to mixed or unclassified contributions.

A small number of very large organizations can affect how a region looks in the data. The findings below describe typical dollars among organizations that report a source, not a complete picture of every non-profit in the region.

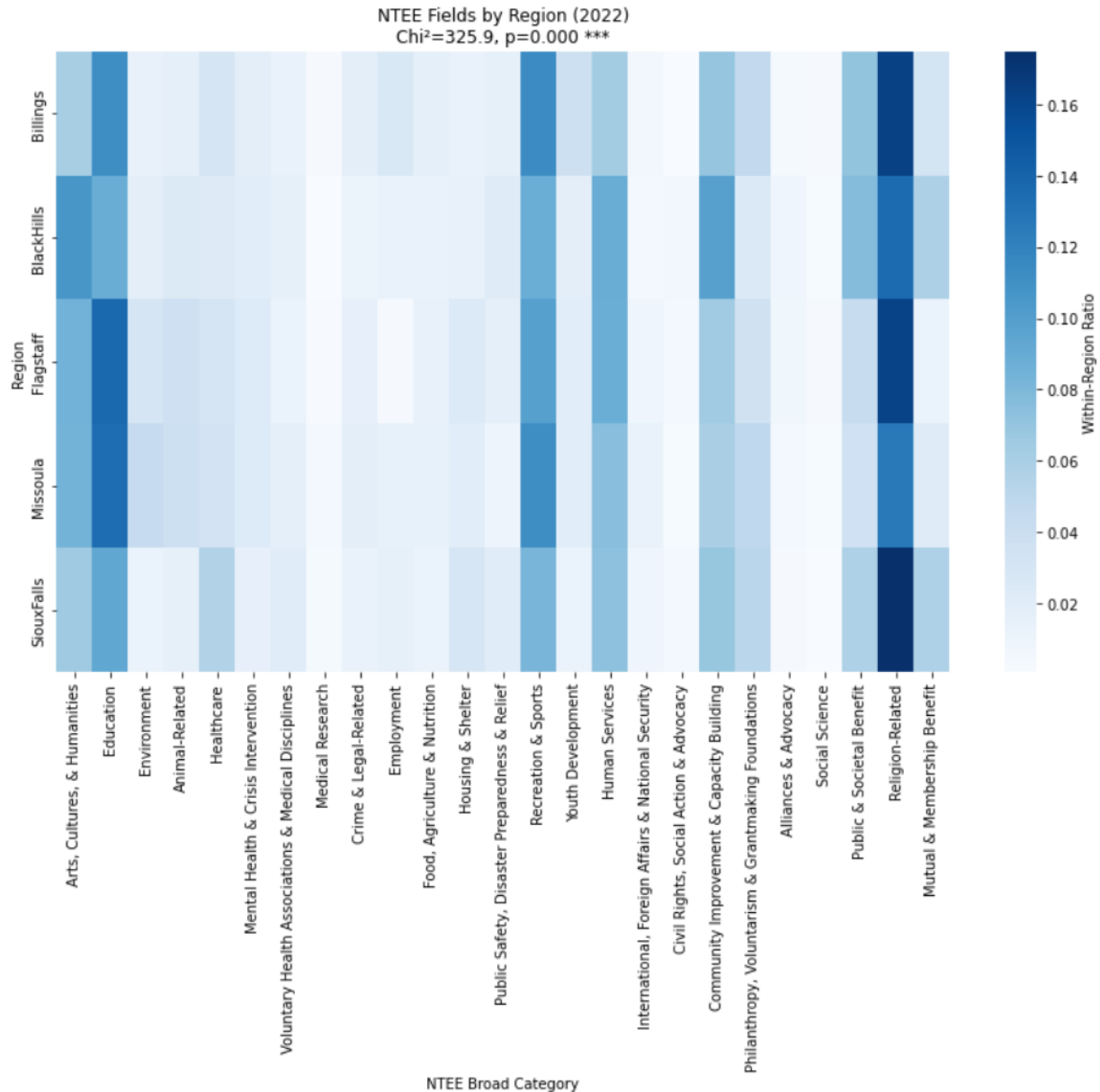
Results of non-profit organization analysis

Form 990 results and IRS-BMF dataset was used to analyze the financial characteristics of non-profit organizations in Black Hills and benchmark regions. Total revenue and assets were used as indicators of organizations' financial performance.

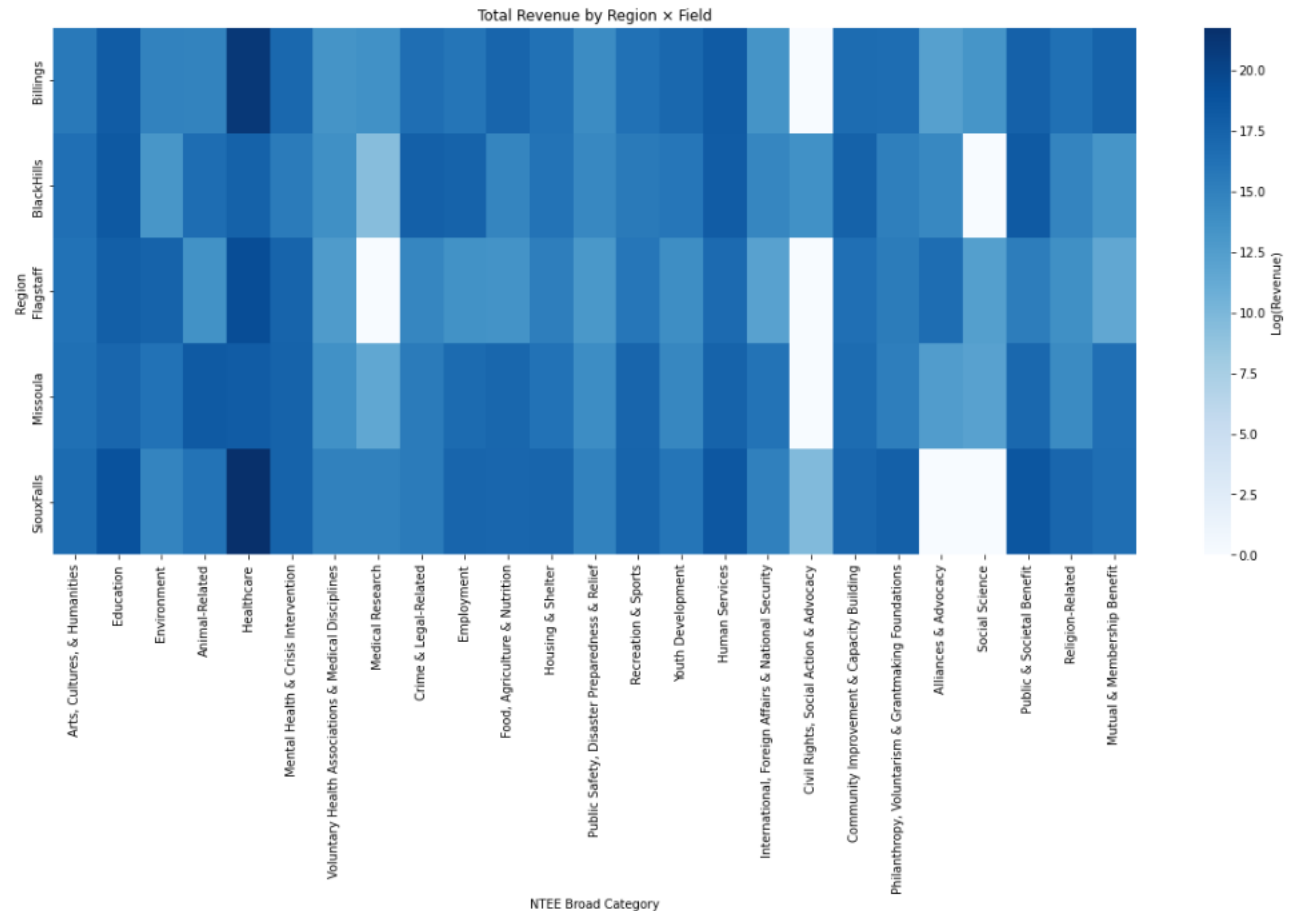
The non-profit densities per 10,000 population calculated on a county-level are not significantly different across the five regions. Although the density varies within regions, there is no particular region that has an extreme nonprofit density statistic compared to their counterparts.



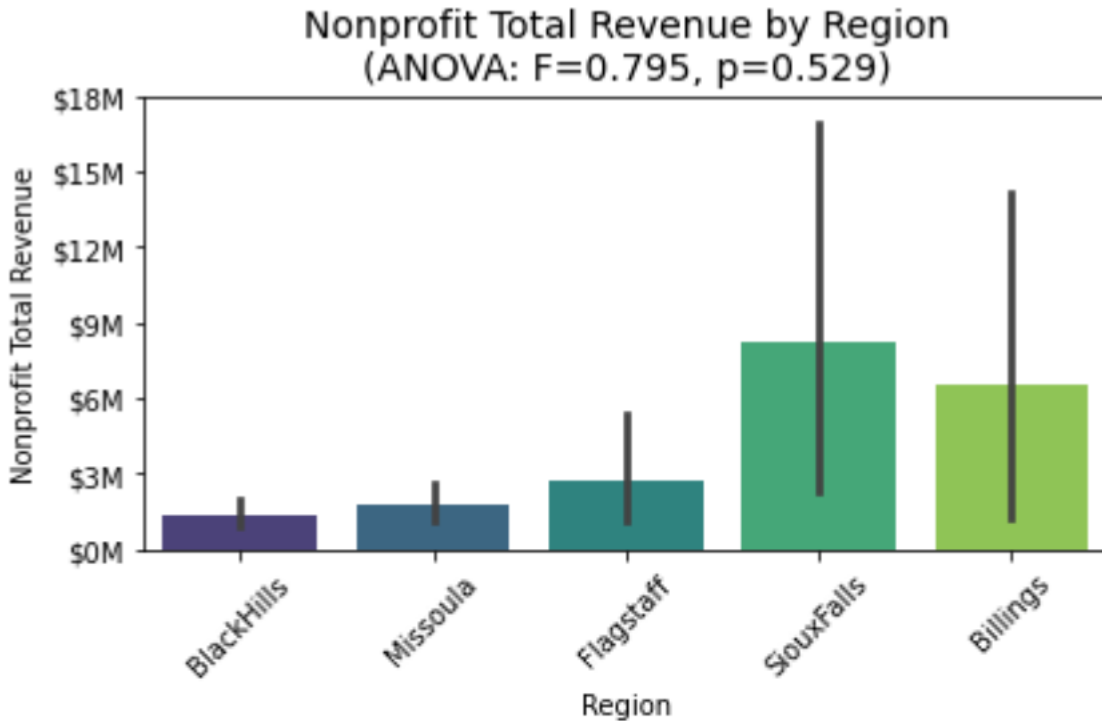
NTEE business field code classifies non-profit organizations based on their activities and main focus. Billings had 26 organizations classified as unidentified, 24 in Black Hills, 11 in Flagstaff, 20 in Missoula, and 56 in Sioux Falls, ranging from 2 - 3% within-region ratios. Billings had 4 missing values, 1 in Missoula, and 6 in Sioux Falls. The heatmap below shows within-region ratios of broad business fields given the total count of non-profit organizations in each region. Billings, Flagstaff, and Sioux Falls show a relatively higher ratio of religious organizations. Sioux Falls shows a slightly higher ratio of healthcare organizations compared to the other four regions.



The second heatmap shows total revenue of non-profit organizations in each field. The total revenue values were logarithmically transformed to compress extremely high outliers and allow analysis focused on lower values. Billings and Sioux Falls show higher total revenue in healthcare sectors. Although religious organizations dominate in terms of numbers, they do not produce high revenue. This is partly due to the IRS not requiring religious organizations to report their revenues.

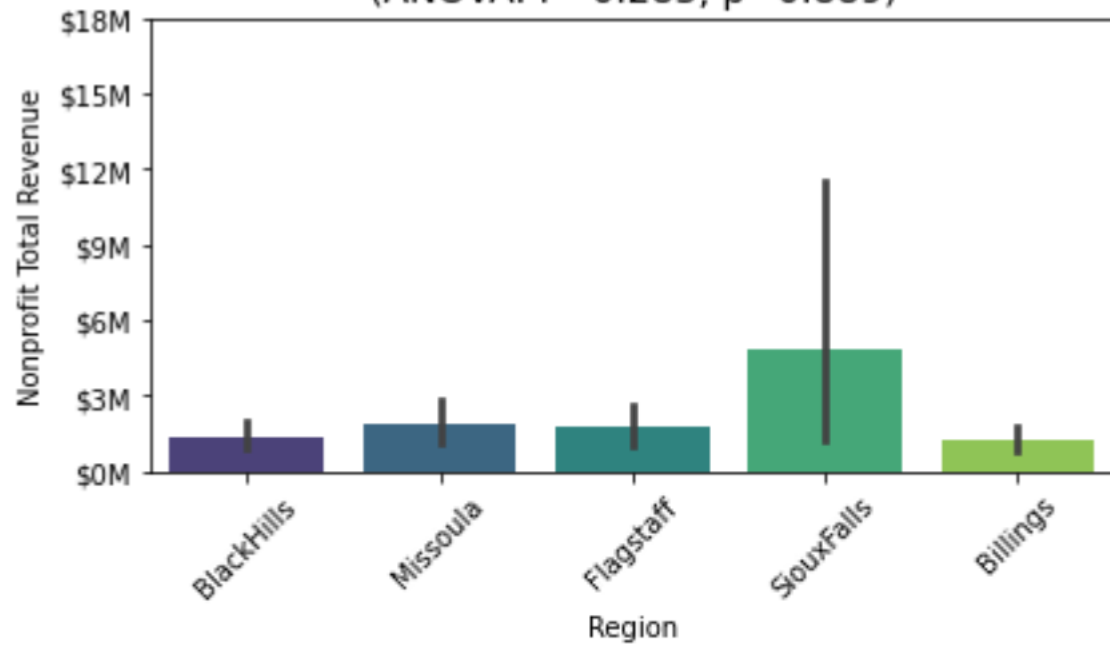


Overall, total revenue of non-profit organizations varies widely within regions. Although Sioux Falls and Billings have high total revenue values on average, the differences between regions are small due to the high variability. The high variabilities in Sioux Falls and Billings are likely due to the revenue gains from healthcare organizations as suggested by the heatmap above. This result suggests that non-profit organizations in Black Hills and benchmark regions generally receive similar amounts of revenue and differences of average total revenue are caused by extreme outliers.



Filtering out hospitals, universities, and political organizations shows a significant decrease in average total revenue of Sioux Falls and Billings. The average total revenue of Billings drops to a similar amount to the average total revenue of Black Hills. The reduction of variability also indicates that the extreme outliers came from the three business fields filtered out in the bar graph below. Sioux Falls continues to have the highest average total revenue.

Nonprofit Total Revenue by Region (Filtered)
(ANOVA: F=0.283, p=0.889)



We compared whether non-profit revenue sources differ between the Black Hills and the benchmark regions. For tax year 2022, the answer is yes. Black Hills differs from at least one benchmark region on several revenue sources. The pattern depends on which revenue source and which benchmark region are being compared.

Each revenue source was analyzed separately using organization-level dollars reported on IRS Form 990-family filings. Comparisons included only organizations that reported more than zero dollars for that specific source in 2022. For each source, we compared the typical reported amount in Black Hills with the typical amount in Billings, Flagstaff, Sioux Falls, and Missoula. The typical amount is the median among organizations that reported that source. Statistical tests used a two-sided permutation test on the difference between those medians. We treated $p < 0.05$ as evidence of a meaningful difference.

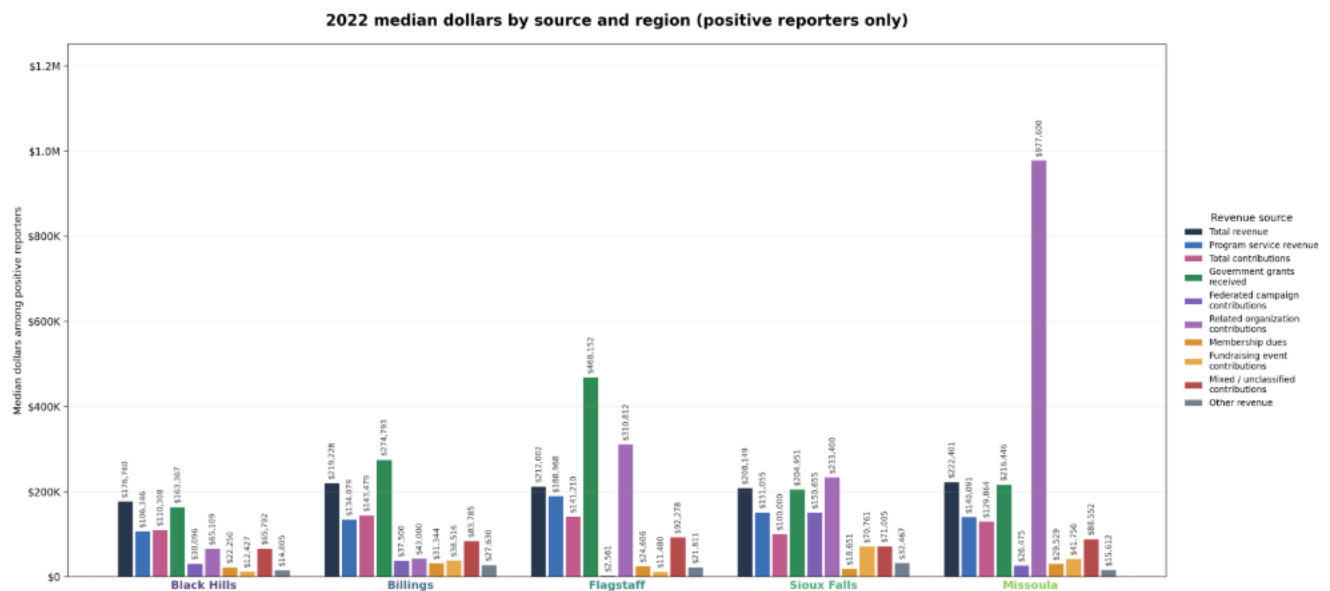


Figure 4. 2022 median reported dollars by revenue source and region. Each panel is one revenue source. Bar height is the typical amount among organizations in that region that reported a positive amount for that source. This figure does not show total dollars summed across all organizations in a region.

How many organizations could we compare?

Only organizations that reported income from a given revenue source were included in that source's comparison. The number of organizations varies widely by source.

The 422 Black Hills records above are all Form 990-family filers with positive total revenue in 2022. Comparisons for Form 990 Part VIII contribution lines (government grants, federated campaigns, related-organization contributions, membership dues, and fundraising events) use a smaller eligible pool: 256 Black Hills organization-years where those lines exist (primarily full Form 990). Reporting rates for those rare sources are shares of that pool, not of all 422 records. Program service revenue uses a slightly different eligible count because 990-EZ filers can report it; see the appendix tables for reporter counts by pair.

For total revenue, total contributions, program service revenue, mixed or unclassified contributions, and other revenue, most pairwise comparisons involved dozens to hundreds of organizations with positive amounts on both sides. Those comparisons are more stable.

Some Part VIII lines are reported by far fewer organizations. In 2022, about 7% of eligible Black Hills filings (the Form 990 subset) reported federated campaign contributions, with 18 organizations showing a positive amount. Some benchmark comparisons for that source involved as few as three to five organizations with positive amounts. Related-organization contributions were reported by about 4% of eligible Black Hills filings, with 10 organizations showing a positive amount. Fundraising event contributions and membership dues fell in between, with moderate sample sizes in some benchmark pairs.

A statistically meaningful difference can still appear when the number of reporting organizations is small. For example, federated campaign contributions differed between Black Hills and Sioux Falls at $p < 0.05$, but that comparison should be read with extra caution because relatively few organizations report that source. Related-organization contributions showed no statistically meaningful pairwise differences, and they also had the smallest reporting groups. An absence of detected difference for those sources may reflect limited data as much as a true lack of gap.

Findings where Black Hills differed from a benchmark region (2022)

Among organizations that reported a positive amount for the source, Black Hills showed a lower typical dollar amount than the benchmark region in each comparison below (permutation $p < 0.05$). Significance is based on the two-sided

permutation test, not on whether the 95% CI on the median gap excludes zero; some significant rows below have a CI that crosses zero.

Revenue source	Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Federated campaign contributions	Sioux Falls	Black Hills lower among reporters; significant	\$30,096	\$150,655	-\$120,559 (95% CI - \$251,704 to - \$58,041)	0.005
Fundraising event contributions	Billings	Black Hills lower among reporters; significant	\$12,427	\$38,516	-\$26,088 (95% CI - \$64,852 to -\$6,639)	0.009
Fundraising event contributions	Missoula	Black Hills lower among reporters; significant	\$12,427	\$41,756	-\$29,329 (95% CI - \$72,681 to \$1,335)	0.010
Fundraising event contributions	Sioux Falls	Black Hills lower among reporters; significant	\$12,427	\$70,761	-\$58,334 (95% CI - \$80,561 to -\$31,388)	< 0.001
Government grants received	Billings	Black Hills lower among reporters; significant	\$163,367	\$274,793	-\$111,426 (95% CI - \$345,341 to \$7,163)	0.043
Government grants received	Flagstaff	Black Hills lower among reporters; significant	\$163,367	\$468,152	-\$304,785 (95% CI - \$817,302 to - \$131,168)	0.003
Mixed / unclassified contributions	Flagstaff	Black Hills lower among reporters; significant	\$65,792	\$92,278	-\$26,486 (95% CI - \$50,596 to -\$2,907)	0.033
Other revenue	Billings	Black Hills lower among reporters; significant	\$14,805	\$27,630	-\$12,825 (95% CI - \$22,923 to -\$1,324)	0.010

Other revenue	Sioux Falls	Black Hills lower among reporters; significant	\$14,805	\$32,467	-\$17,662 (95% CI - \$25,196 to -\$6,938)	0.001
Program service revenue	Flagstaff	Black Hills lower among reporters; significant	\$106,346	\$188,968	-\$82,622 (95% CI - \$188,223 to \$1,513)	0.038

For total revenue and total contributions, none of the four Black Hills vs benchmark comparisons reached $p < 0.05$. The appendix tables list every source and benchmark pair, including comparisons that were not statistically meaningful at $p < 0.05$.

Participation and dollar amounts can tell different stories. Government grants are a useful example: a larger share of eligible Black Hills organizations reported the source (about 44%) than in Billings (about 34%), yet the typical grant amount among Black Hills reporters was lower (\$163,367 versus \$274,793, $p = 0.043$). A region can have more organizations using a revenue source while still showing a lower typical dollar amount among the organizations that use it. Reporting rates by source and region are available in the supporting analysis files referenced in the appendix.

The appendix lists the full set of p-values and confidence intervals from the client presentation, organized by revenue source, along with matching source-by-source bar charts (Figures 5–14).

For the Black Hills Area Community Foundation, these results suggest that many local non-profits may operate in a funding environment that differs from the benchmark regions. If this pattern aligns with local experience, it could affect how organizations plan for sustainability, pursue grants, and communicate funding needs to donors.

These findings describe 2022 reported revenue patterns by region, not causal explanations. Differences in organization type, local service needs, government funding, tourism-related activity, or the presence of large institutions could contribute but would require context beyond these filings. They also do not apply to every comparable organization in the region, and they do not separate individual giving from institutional giving. Interpret them together with reporting counts in the appendix and the sample-size discussion above.

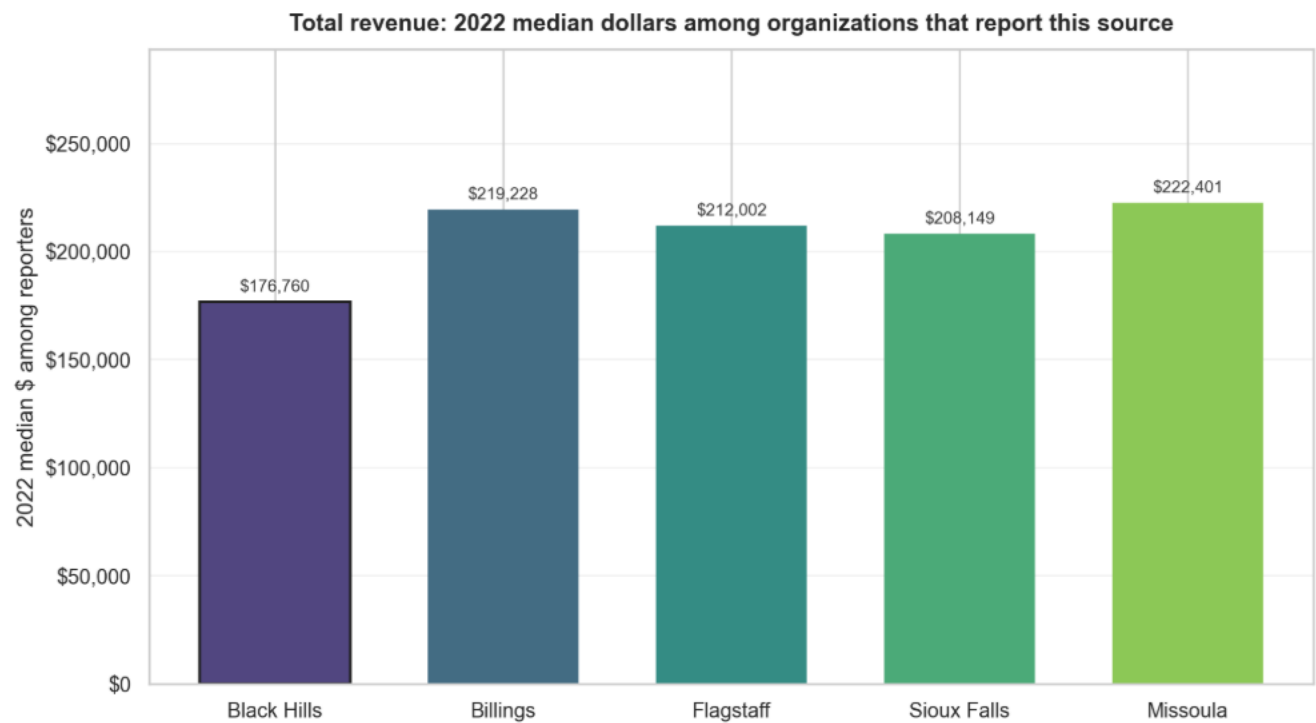
6. Conclusions

Write when finished

Appendix: Non-profit revenue source comparison details

The tables below list all p-values shown in the client presentation, organized by revenue source. Figures 5–14 show the typical reported dollar amount by region for each revenue source (2022, among organizations that reported a positive amount for that source). These comparisons use organization-level reported dollars. Rare revenue sources have fewer reporting organizations; see the sample-size discussion in the results section above.

Total revenue

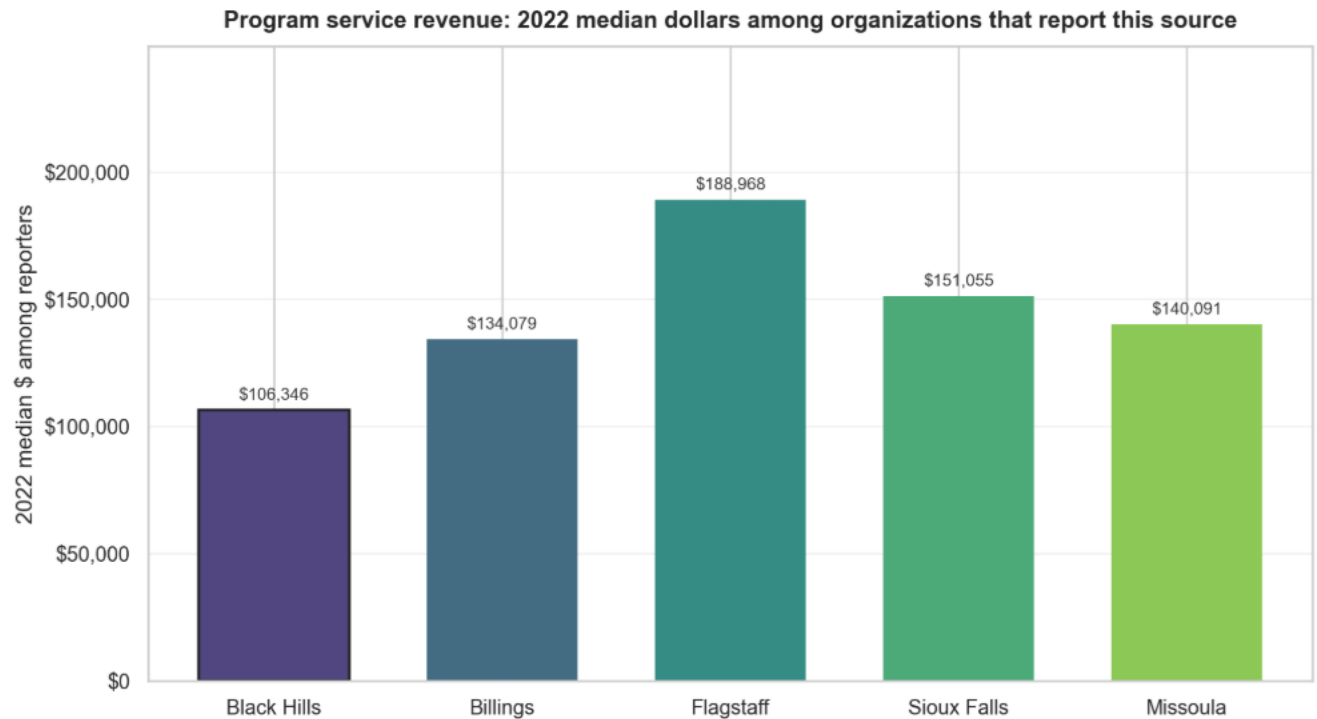


Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; not significant	\$176,760	\$219,228	-\$42,468 (95% CI - \$125,394 to +\$33,850)	0.269

Flagstaff	Black Hills lower among reporters; not significant	\$176,760	\$212,002	-\$35,242 (95% CI - \$149,323 to +\$35,127)	0.366
Sioux Falls	Black Hills lower among reporters; not significant	\$176,760	\$208,149	-\$31,390 (95% CI - \$84,901 to +\$26,666)	0.246
Missoula	Black Hills lower among reporters; not significant	\$176,760	\$222,401	-\$45,642 (95% CI - \$126,653 to +\$32,158)	0.242

Figure 5. Total revenue (2022): typical reported dollars by region among organizations that reported positive total revenue.

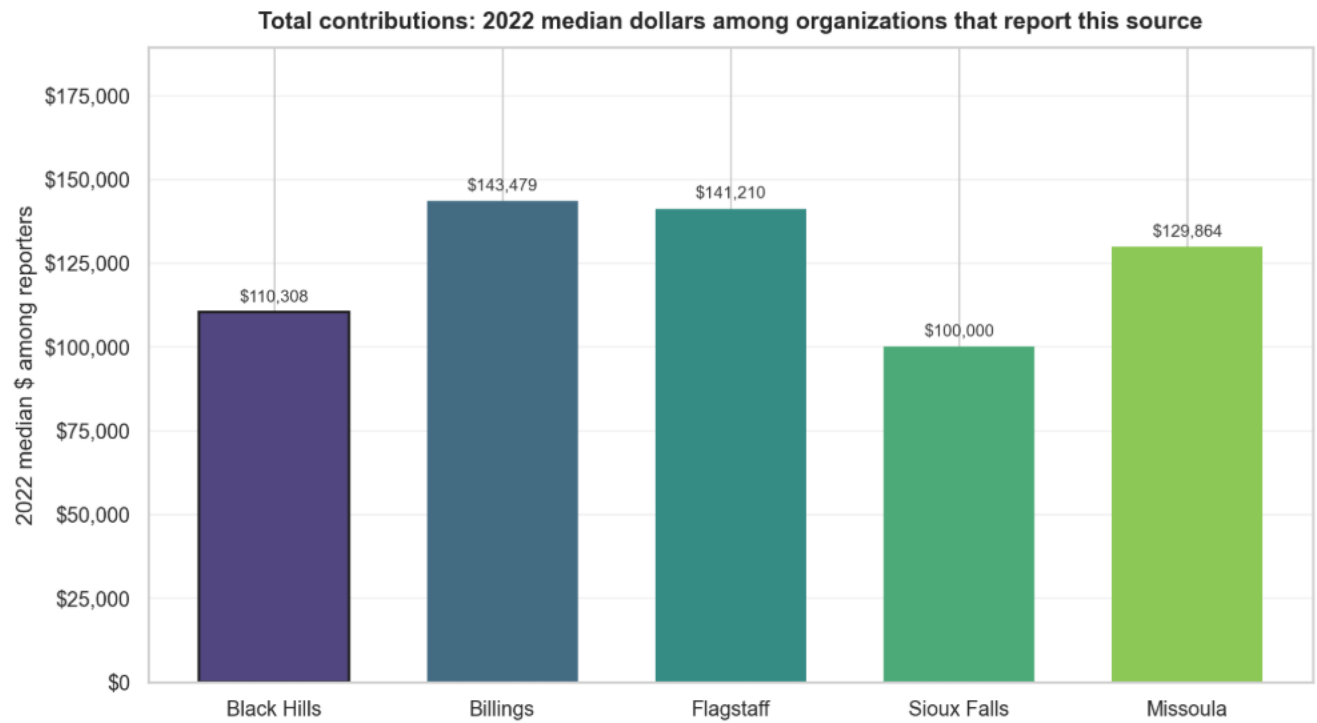
Program service revenue



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; not significant	\$106,346	\$134,079	-\$27,733 (95% CI - \$96,511 to +\$25,161)	0.292
Flagstaff	Black Hills lower among reporters; significant	\$106,346	\$188,968	-\$82,622 (95% CI - \$188,223 to +\$1,513)	0.038
Sioux Falls	Black Hills lower among reporters; not significant	\$106,346	\$151,055	-\$44,709 (95% CI - \$114,440 to +\$6,707)	0.089
Missoula	Black Hills lower among reporters; not significant	\$106,346	\$140,091	-\$33,745 (95% CI - \$91,143 to +\$18,223)	0.175

Figure 6. Program service revenue (2022): typical reported dollars by region among organizations that reported positive program service revenue.

Total contributions



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; not significant	\$110,308	\$143,479	-\$33,171 (95% CI - \$85,280 to +\$16,324)	0.161
Flagstaff	Black Hills lower among reporters; not significant	\$110,308	\$141,210	-\$30,902 (95% CI - \$95,512 to +\$14,239)	0.224
Sioux Falls	Black Hills higher among reporters; not significant	\$110,308	\$100,000	+\$10,308 (95% CI - \$20,952 to +\$41,476)	0.448
Missoula	Black Hills lower among reporters; not significant	\$110,308	\$129,864	-\$19,556 (95% CI - \$95,775 to +\$26,776)	0.309

Figure 7. Total contributions (2022): typical reported dollars by region among organizations that reported positive total contributions.

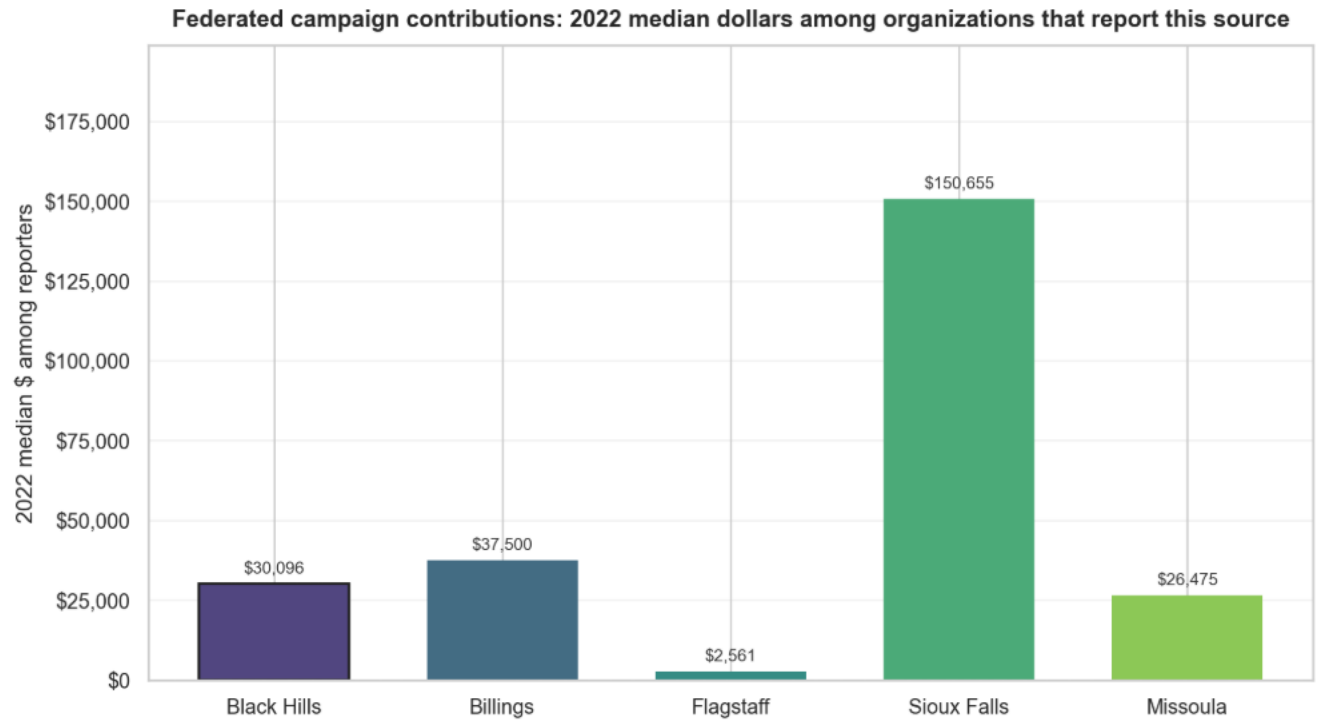
Government grants received



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; significant	\$163,367	\$274,793	-\$111,426 (95% CI -\$345,341 to +\$7,163)	0.043
Flagstaff	Black Hills lower among reporters; significant	\$163,367	\$468,152	-\$304,785 (95% CI -\$817,302 to -\$131,168)	0.003
Sioux Falls	Black Hills lower among reporters; not significant	\$163,367	\$204,951	-\$41,584 (95% CI -\$156,414 to +\$89,266)	0.507
Missoula	Black Hills lower among reporters; not significant	\$163,367	\$216,446	-\$53,078 (95% CI -\$292,032 to +\$26,850)	0.197

Figure 8. Government grants received (2022): typical reported dollars by region among organizations that reported positive government grants.

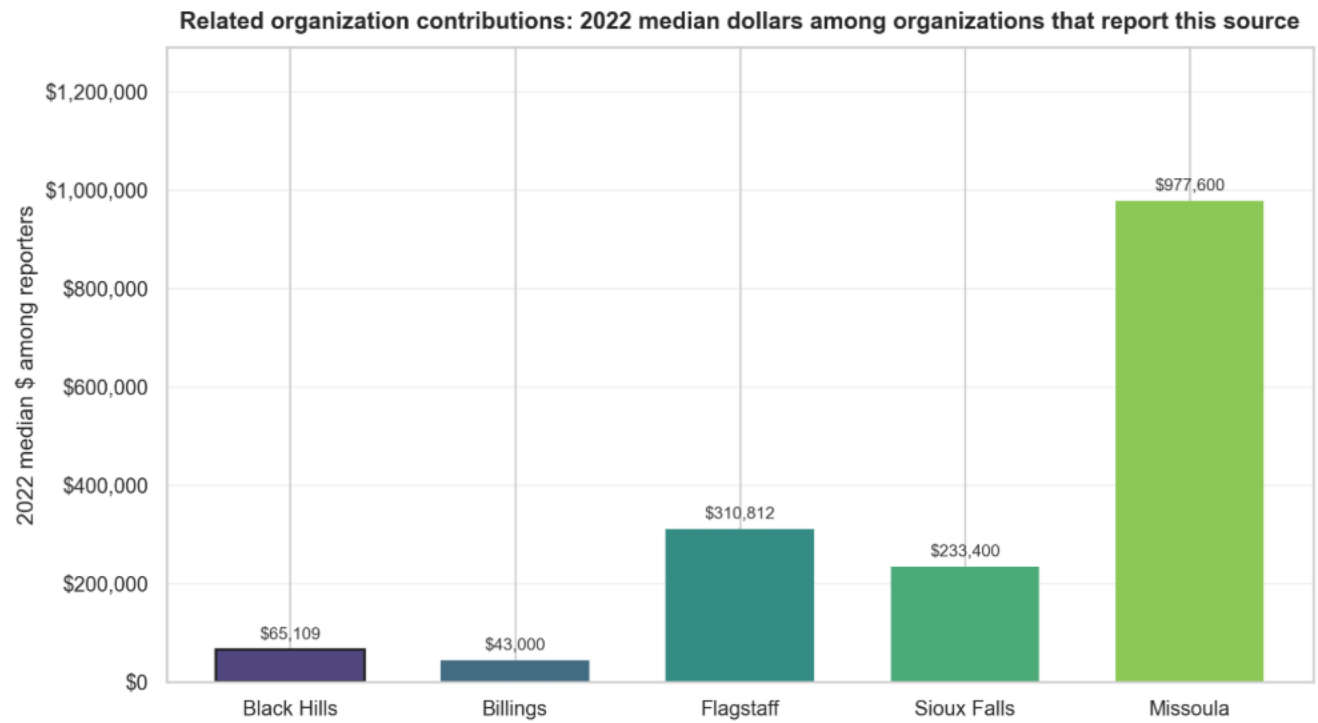
Federated campaign contributions



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; not significant	\$30,096	\$37,500	-\$7,404 (95% CI - \$41,337 to +\$36,319)	0.688
Flagstaff	Black Hills higher among reporters; not significant	\$30,096	\$2,561	+\$27,535 (95% CI - \$50,214 to +\$38,331)	0.179
Sioux Falls	Black Hills lower among reporters; significant	\$30,096	\$150,655	-\$120,559 (95% CI -\$251,704 to -\$58,041)	0.005
Missoula	Black Hills higher among reporters; not significant	\$30,096	\$26,475	+\$3,621 (95% CI - \$374,904 to +\$29,973)	0.819

Figure 9. Federated campaign contributions (2022): typical reported dollars by region among organizations that reported positive federated campaign contributions.

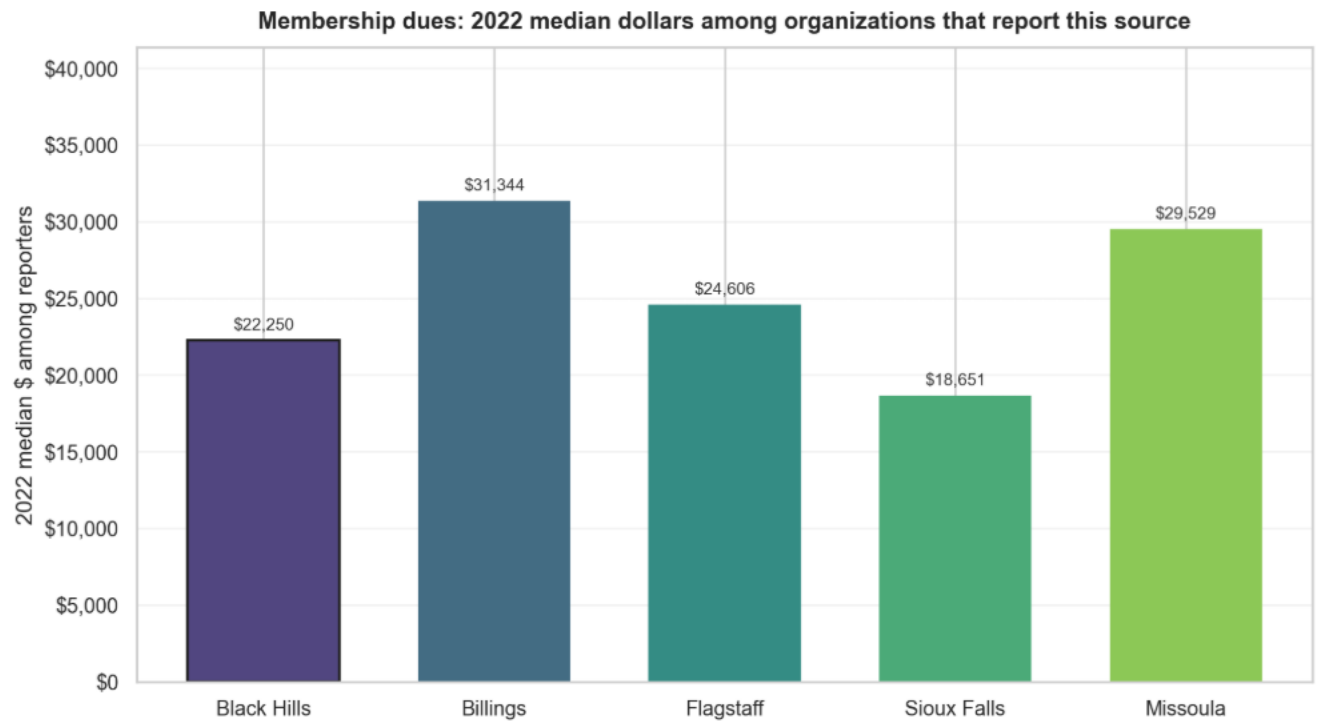
Related organization contributions



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills higher among reporters; not significant	\$65,109	\$43,000	+\$22,109 (95% CI - \$205,000 to +\$1,417,018)	0.627
Flagstaff	Black Hills lower among reporters; not significant	\$65,109	\$310,812	-\$245,703 (95% CI - \$573,240 to +\$1,112,196)	0.126
Sioux Falls	Black Hills lower among reporters; not significant	\$65,109	\$233,400	-\$168,291 (95% CI - \$1,493,390 to +\$1,137,398)	0.203
Missoula	Black Hills lower among reporters; not significant	\$65,109	\$977,600	-\$912,491 (95% CI - \$3,371,000 to +\$150,595)	0.173

Figure 10. Related organization contributions (2022): typical reported dollars by region among organizations that reported positive related-organization contributions.

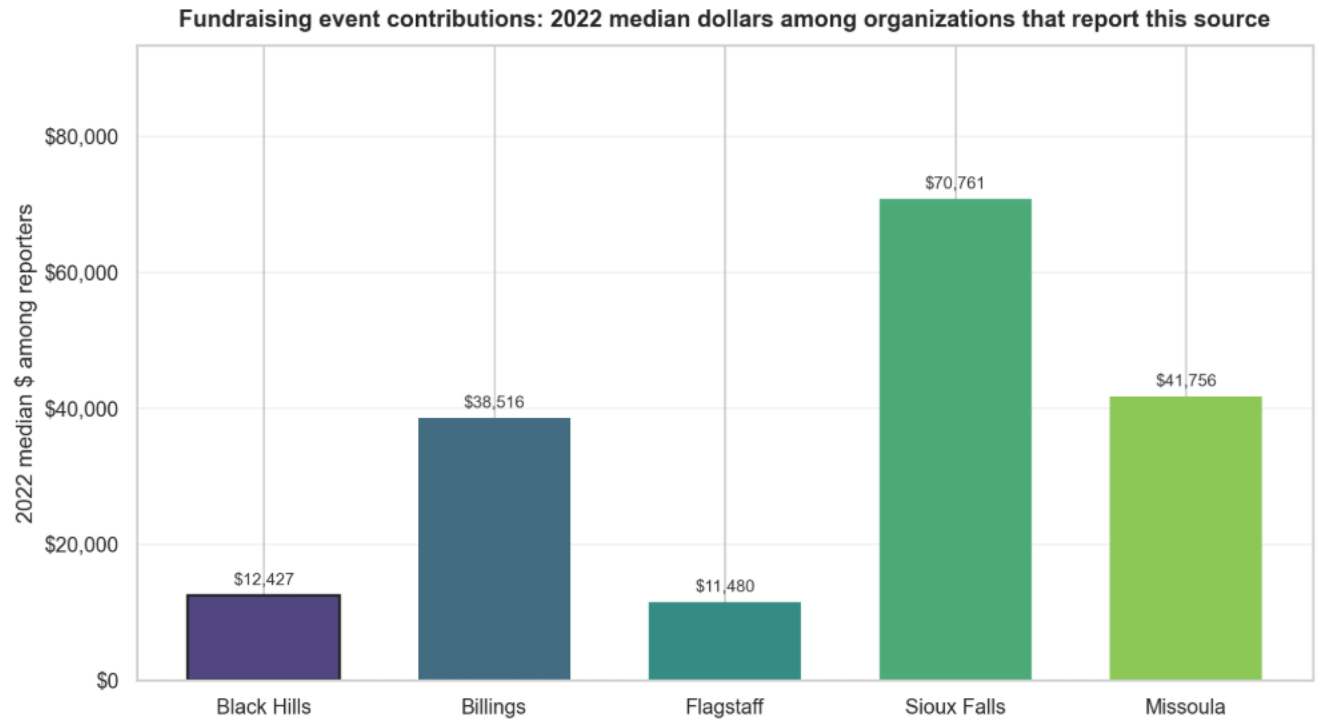
Membership dues



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; not significant	\$22,250	\$31,344	-\$9,093 (95% CI - \$51,928 to +\$10,338)	0.342
Flagstaff	Black Hills lower among reporters; not significant	\$22,250	\$24,606	-\$2,355 (95% CI - \$167,924 to +\$21,272)	0.832
Sioux Falls	Black Hills higher among reporters; not significant	\$22,250	\$18,651	+\$3,600 (95% CI - \$54,010 to +\$23,295)	0.540
Missoula	Black Hills lower among reporters; not significant	\$22,250	\$29,529	-\$7,278 (95% CI - \$20,952 to +\$13,858)	0.601

Figure 11. Membership dues (2022): typical reported dollars by region among organizations that reported positive membership dues.

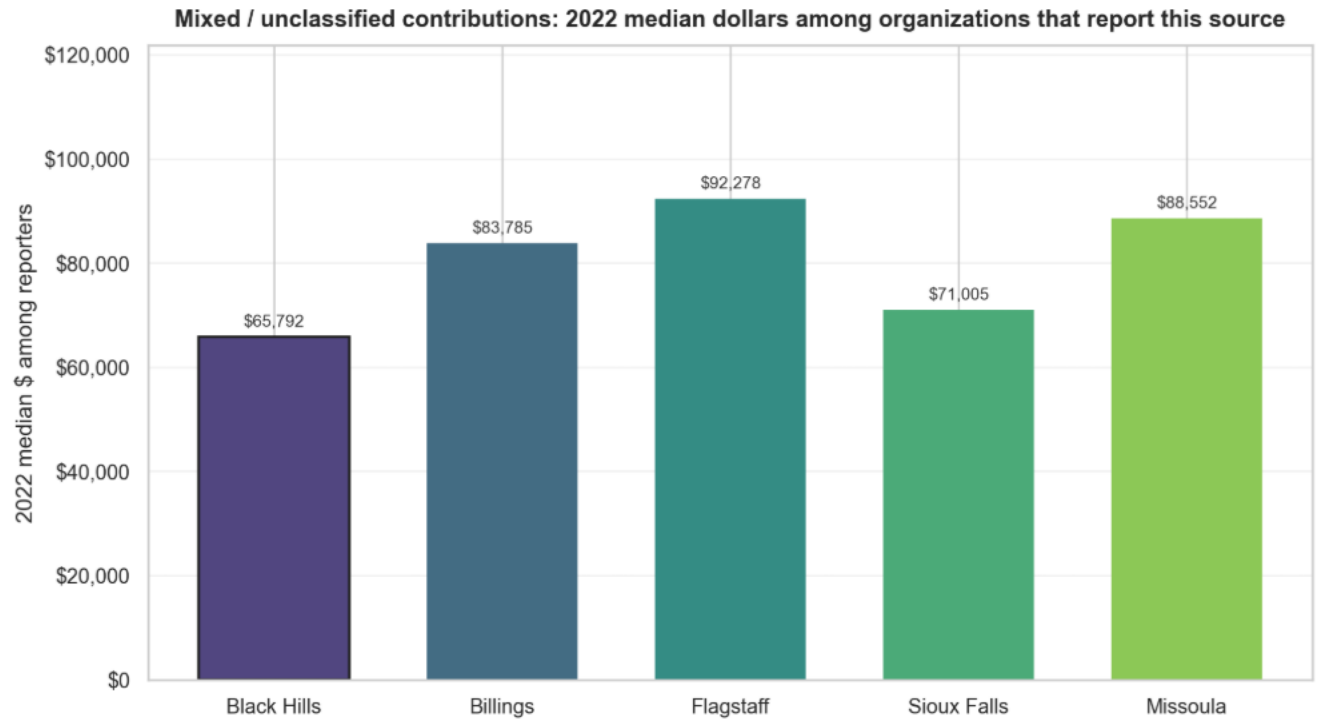
Fundraising event contributions



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; significant	\$12,427	\$38,516	-\$26,088 (95% CI - \$64,852 to - \$6,639)	0.009
Flagstaff	Black Hills higher among reporters; not significant	\$12,427	\$11,480	+\$947 (95% CI - \$9,712 to +\$13,322)	0.849
Sioux Falls	Black Hills lower among reporters; significant	\$12,427	\$70,761	-\$58,334 (95% CI - \$80,561 to - \$31,388)	< 0.001
Missoula	Black Hills lower among reporters; significant	\$12,427	\$41,756	-\$29,329 (95% CI - \$72,681 to +\$1,335)	0.010

Figure 12. Fundraising event contributions (2022): typical reported dollars by region among organizations that reported positive fundraising event contributions.

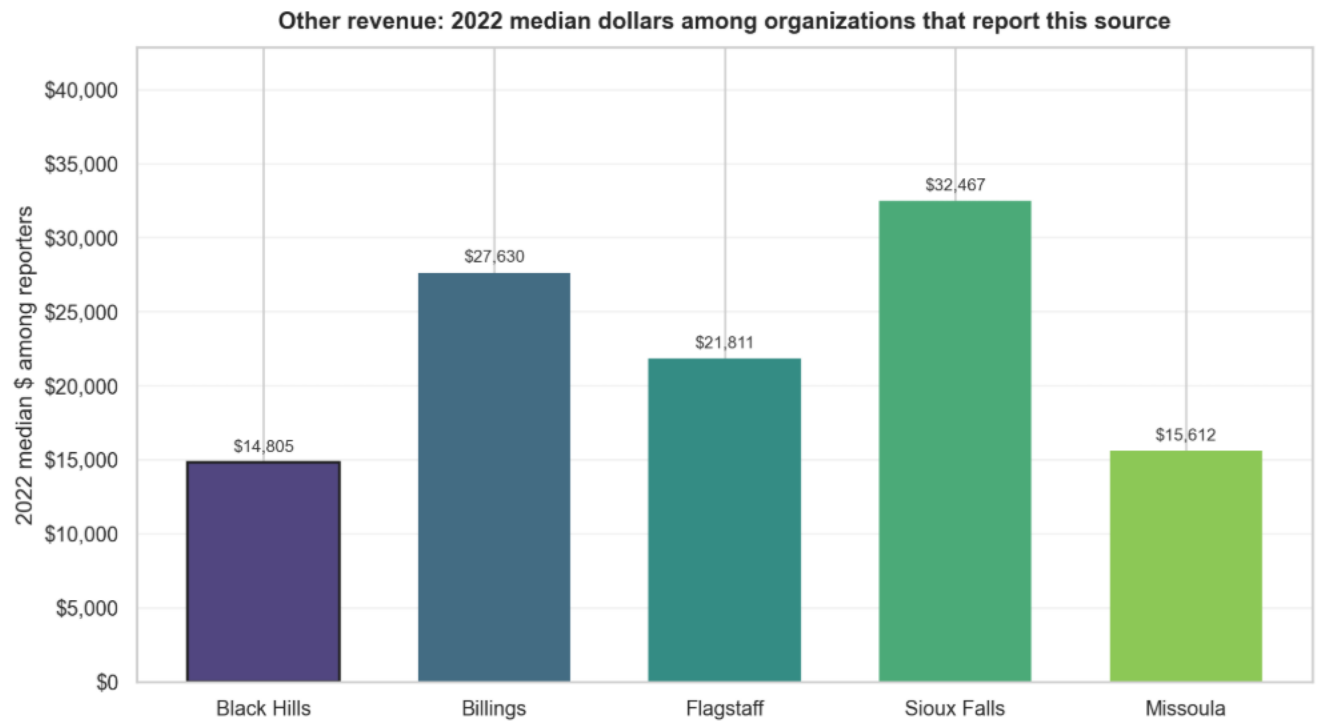
Mixed / unclassified contributions



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; not significant	\$65,792	\$83,785	-\$17,992 (95% CI - \$49,754 to +\$8,827)	0.157
Flagstaff	Black Hills lower among reporters; significant	\$65,792	\$92,278	-\$26,486 (95% CI - \$50,596 to - \$2,907)	0.033
Sioux Falls	Black Hills lower among reporters; not significant	\$65,792	\$71,005	-\$5,212 (95% CI - \$29,255 to +\$15,095)	0.619
Missoula	Black Hills lower among reporters; not significant	\$65,792	\$88,552	-\$22,760 (95% CI - \$58,330 to +\$11,256)	0.099

Figure 13. Mixed or unclassified contributions (2022): typical reported dollars by region among organizations that reported positive mixed or unclassified contributions.

Other revenue



Benchmark region	Direction	Black Hills median	Benchmark median	Median gap (95% CI)	p-value
Billings	Black Hills lower among reporters; significant	\$14,805	\$27,630	-\$12,825 (95% CI - \$22,923 to - \$1,324)	0.010
Flagstaff	Black Hills lower among reporters; not significant	\$14,805	\$21,811	-\$7,006 (95% CI - \$27,475 to +\$6,558)	0.292
Sioux Falls	Black Hills lower among reporters; significant	\$14,805	\$32,467	-\$17,662 (95% CI - \$25,196 to - \$6,938)	0.001
Missoula	Black Hills lower among reporters; not significant	\$14,805	\$15,612	-\$807 (95% CI - \$8,600 to +\$7,563)	0.697

Figure 14. Other revenue (2022): typical reported dollars by region among organizations that reported positive other revenue.

The full client presentation and supporting analysis files include reporting rates and additional source-by-source interpretation.